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SLICE: Sequential Likelihood-Free Inference via Conformal E-Values

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Abstract

Many scientific models can be simulated but have intractable likelihoods, making frequentist uncertainty quantification difficult. We construct exact anytime-valid confidence sequences for the parameters of such models. At each observation, fresh simulations from each candidate null conformalize a positive predictable score into a conditional e-value; their product is a test martingale, which we invert over the parameter. Validity holds for any such score, whatever its accuracy. We show that the oracle likelihood-ratio score maximizes expected log growth, which motivates approximating it with an amortized neural ratio estimator. On synthetic benchmarks, an audit of model routing in an LLM API, and microlensing data, neural scores remain competitive with exact likelihood ratios and application-specific scores.

1 INTRODUCTION

Many scientific models are easiest to express as simulators. Given a parameter θ, a computer program can generate synthetic observations $Y \sim P_\theta$, but evaluating the likelihood $f_\theta(y)$ may be impossible or prohibitively expensive. This setting, often called likelihood-free or simulation-based inference, occurs in population genetics, ecology, epidemiology, cosmology or climate science [6]. Inference replaces likelihood evaluation with repeated simulation, using methods such as neural density estimation or classification-based likelihood-ratio estimation. These methods can be effective, but their uncertainty quantification is usually approximate or designed for a prespecified sample size [21, 44, 30, 7, 8].

We study *anytime-valid frequentist uncertainty quantification* in likelihood-free models. For observations $Y_1, Y_2, \ldots \overset{\text{i.i.d.}}{\sim} P_{\theta_*}$ arriving sequentially, we seek a sequence of random sets $(C_{n,\alpha})_{n\geq 1}$ forming a $(1-\alpha)$ confidence sequence (CS) [27, 23], i.e., satisfying

$$\mathbb{P}_{\theta_*}(\theta_* \in C_{n,\alpha} \text{ for all } n \geq 1) \geq 1-\alpha \qquad (1)$$

for every data generating parameter θ_*. Unlike a collection of fixed-sample confidence sets, a CS remains valid under continuous monitoring and at data-dependent stopping times. Existing frequentist approaches to likelihood-free inference primarily construct and calibrate fixed-sample tests or confidence regions [7, 8]. Obtaining the time-uniform guarantee in (1) requires a sequentially valid construction. When likelihoods are available, confidence sequences can be obtained, via Ville's inequality [46], by inverting likelihood-ratio test martingales [38, 37]. In implicit models the ratios cannot be evaluated, and simply substituting a neural estimate generally destroys the martingale property on which the guarantee relies.

We develop **SLICE**: Sequential Likelihood-Free Inference via Conformal E-Values. SLICE turns amortized neural ratio estimates into anytime-valid confidence sequences. It builds on normalized-score conformal e-values [15, 1, 47] and their sequential or simulation-based extensions [17, 10].

For each candidate parameter, we evaluate the incoming observation and fresh simulations from the candidate model using the same positive score function, chosen using only past information. Dividing the observed score by the average score across the observation and its references gives a *normalized-score e-value*. Multiplying these factors over time provides a test martingale, whose inversion yields a confidence sequence. Coverage holds for any positive predictable score and any number of reference observations.

The score nevertheless matters for efficiency. We assess it through expected log growth, a measure of how quickly evidence accumulates against false candidates. Specializing the log-optimality results of Koning [24], we identify the likelihood ratio as an optimal score within the normalized-score construction. A classification-based proof connects this result to contrastive learning [45, 11] and bounds the efficiency lost

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to finitely many references. In practice, we approximate the optimal score with an amortized neural ratio estimator [21, 30].

We evaluate SLICE on Gaussian and g-and-k location models and the SLCP benchmark, where neural scores yield confidence sequences close to conformalized likelihood ratios. We then apply it to a semi-synthetic audit of model routing in an LLM API, using real model replies, and to inference of the stellar initial mass function from OGLE-IV microlensing events. In both, neural score confidence sequences perform comparably to those using application-specific scores.

2 RELATED WORK

Simulation-based conformal e-values. Normalizing scores over an exchangeable batch yields conformal e-values [47]. Closely related, Dombowsky et al. [10] develop simulation-based normalized-score e-values, confidence-set inversion, and sequential products, emphasizing Besag–Clifford MCMC while also treating independent null simulation. Our oracle-score result specializes the log-optimality of normalized likelihood ratios under permutation invariance [24], and our finite-reference bound specializes the log-growth comparison of Dombowsky et al. [10]. SLICE builds on these by combining direct null simulation with predictable alternatives and amortized neural ratio scores for sequential likelihood-free parameter inference.

Frequentist simulation-based inference. ACORE and LF2I construct fixed-sample tests and confidence regions using simulation-based calibration [7, 8]. WALDO similarly calibrates a Wald-type statistic built from prediction algorithms or posterior estimators [29]. TRUST and TRUST++ use conformal calibration of statistical confidence sets [3]. These approaches address fixed-sample inference. SLICE instead uses fresh simulations from each candidate null to normalize the score of every incoming observation, then inverts the resulting test martingales. Under the stated sampling conditions, this gives time-uniform coverage for any positive predictable score and reference-batch size, without estimating critical values.

Neural ratio estimation. Our scores build on classification-based amortized ratio estimation and contrastive learning [21, 11, 30]. We use a likelihood-to-evidence network to construct ratios between a predictable alternative and each candidate null. The resulting neural score need not be an exact likelihood ratio or itself a valid e-value: conditional exchangeability and score normalization supply validity, while score quality affects efficiency. This permits approximate neural models to support exact sequential inference. Further connections to posterior calibration, density-ratio estimation, and sequential testing are discussed in Section A.

Concurrent work. After the submission of an earlier version of this manuscript, an independent concurrent work by Forré and Brenner [14] appeared on arXiv, studying likelihood-free anytime-valid hypothesis testing. Both works use the normalized-score construction, building on earlier literature [14, cf. Lemma 3.1 and Eqs. (56)–(58)], and establish closely related oracle optimality and finite-reference log-growth bounds (our Theorem 4.2; [14, Proposition 3.2]). Forré and Brenner [14] additionally provide theoretical guarantees for approximate growth optimality of empirically trained scores. Our emphasis is on parameter confidence sequences obtained by test inversion, parameter-amortized neural ratio scores with predictable adaptive alternatives, and extensive empirical evaluation on synthetic benchmarks and applications.

3 PRELIMINARIES

Notation and statistical setting. Let $\{P_\theta : \theta \in \Theta\}$ be a family of probability distributions on $(\mathcal{Y}, \mathcal{B})$, dominated by a common σ-finite measure μ, with densities $f_\theta = \mathrm{d}P_\theta/\mathrm{d}\mu$. The observations are $Y_1, Y_2, \ldots \overset{\text{i.i.d.}}{\sim} P_{\theta_*}$, where $\theta_* \in \Theta$ is fixed and unknown. For each $\theta \in \Theta$, write $\mathbb{P}_\theta$ for the joint law of the observations and all auxiliary randomness under $H_\theta : \theta_* = \theta$, and $\mathbb{E}_\theta$ for expectation under this law.

Let $(\mathcal{F}_n)_{n\geq 0}$ be the filtration generated by the observations and auxiliary information revealed through time n which, in the context of this paper, will include simulator draws. The initial sigma-field $\mathcal{F}_0$ contains pretraining and initialization information. We assume that Y_i is independent of $\mathcal{F}_{i-1}$ for every $i \geq 1$.

Finally, let $r_{\theta_1,\theta_0}(y) = f_{\theta_1}(y)/f_{\theta_0}(y)$ be the likelihood ratio, where $f_{\theta_0}(y) > 0$.

3.1 Test martingales and confidence sequences

For a candidate parameter θ, a nonnegative, adapted, integrable process $(M_n^{(\theta)})_{n\geq 0}$, initialized at $M_0^{(\theta)} = 1$, is a *test supermartingale* under H_θ if

$$\mathbb{E}_\theta\left[M_n^{(\theta)} \mid \mathcal{F}_{n-1}\right] \leq M_{n-1}^{(\theta)} \quad \mathbb{P}_\theta\text{-a.s.}, \qquad n \geq 1.$$

It is a *test martingale* when equality holds [40].

A useful construction multiplies *conditional e-values*: nonnegative, $\mathcal{F}_n$-measurable factors $E_n^{(\theta)}$ satisfying

$$\mathbb{E}_\theta\left[E_n^{(\theta)} \mid \mathcal{F}_{n-1}\right] \leq 1. \tag{2}$$

Their product $M_n^{(\theta)} = \prod_{i=1}^n E_i^{(\theta)}$ is a test supermartingale, and is a test martingale if each factor has conditional expectation one.

For every $\alpha \in (0,1)$, Ville's inequality [46] gives

$$\mathbb{P}_\theta\left(\sup_{n\geq 0} M_n^{(\theta)} \geq \alpha^{-1}\right) \leq \alpha. \tag{3}$$

Consequently, if $(M_n^{(\theta)})_{n\geq 0}$ is a test supermartingale under $\mathbb{P}_\theta$ for every $\theta \in \Theta$, then $C_{n,\alpha} = \{\theta \in \Theta : M_n^{(\theta)} < \alpha^{-1}\}$, $n \geq 1$, forms a $(1-\alpha)$ confidence sequence [27, 23], that is, satisfies (1) for every $\theta_* \in \Theta$. See Section B.1 for the coverage argument.

Likelihood-ratio test martingale. For a fixed alternative $\vartheta \in \Theta$, taking $E_i^{(\theta)} = r_{\vartheta,\theta}(Y_i)$ gives

$$M_n^{(\theta)} = \prod_{i=1}^n r_{\vartheta,\theta}(Y_i), \qquad M_0^{(\theta)} = 1. \tag{4}$$

Under H_θ, each factor has conditional expectation one given $\mathcal{F}_{i-1}$, so this process is a test martingale [37].

The choice $\vartheta = \theta_*$ gives the *oracle likelihood-ratio martingale*, whose expected log growth is

$$\frac{1}{n}\mathbb{E}_{\theta_*}\left[\log M_n^{(\theta)}\right] = \mathrm{KL}(P_{\theta_*} \| P_\theta), \qquad n \geq 1,$$

provided this divergence is finite. This oracle choice maximizes expected log growth under $\mathbb{P}_{\theta_*}$ among all $(\mathcal{F}_n)$-adapted test martingales under $\mathbb{P}_\theta$ initialized at one [37, Theorem 7.11]. This benchmark requires the unknown parameter θ_* and likelihood ratios that are unavailable in our setting. Substituting learned ratios generally loses the conditional expectation property (2), motivating the normalization developed in Section 4.1.

3.2 Neural ratio estimation

Neural ratio estimation (NRE) learns likelihood ratios from simulator draws by classification, without evaluating the underlying densities [5, 21]. In amortized NRE, a single network $s_\phi(\theta, y)$ takes a simulator parameter value θ and an observation y as inputs; ϕ denotes its trainable weights. Given a training density q over Θ, define the marginal density $m(y) = \int_\Theta f_\vartheta(y) q(\vartheta)\,\mathrm{d}\vartheta$.

We use NRE-C [30], which trains a multiclass classifier to identify which candidate parameter generated a simulated observation, with an additional class in which the observation is independent of all candidates. Its ideal population target is the log likelihood-to-evidence ratio

$$s^\star(\theta, y) = \log \frac{f_\theta(y)}{m(y)}. \tag{5}$$

Consequently, differences of network outputs provide likelihood-ratio estimates:

$$\exp\{s_\phi(\theta_1, y) - s_\phi(\theta_0, y)\} \approx r_{\theta_1,\theta_0}(y) = \frac{f_{\theta_1}(y)}{f_{\theta_0}(y)},$$

since the marginal density cancels. This is the construction used for the neural scores in Section 4.3. The training objectives and alternative estimators, including NRE-A and NRE-B, are discussed in Section B.2; implementation details are given in Section E.2.

4 SEQUENTIAL UNCERTAINTY QUANTIFICATION FOR LIKELIHOOD-FREE MODELS

We first construct a likelihood-free test martingale by normalizing a predictable score using fresh null simulations, and invert these martingales to obtain confidence sequences. We then characterize the score that maximizes expected log growth and use this characterization to motivate a practical implementation based on amortized neural ratio estimation.

4.1 A likelihood-free test martingale

Fix a candidate $\theta \in \Theta$ and an integer $K \geq 1$. At time i, choose a positive measurable score

$$T_i^{(\theta)} : \mathcal{Y} \to (0, \infty) \tag{6}$$

using only $\mathcal{F}_{i-1}$, before observing Y_i or its reference simulations. We call such a score *predictable*.

Generate a reference batch

$$\widetilde{Y}_{i1}^{(\theta)}, \ldots, \widetilde{Y}_{iK}^{(\theta)} \overset{\text{i.i.d.}}{\sim} P_\theta, \tag{7}$$

independently of $(\mathcal{F}_{i-1}, Y_i)$. All reference batches generated through time i, across all candidate parameters, are included in the auxiliary randomness recorded in $\mathcal{F}_i$. Under H_θ, conditional on $\mathcal{F}_{i-1}$, the observation and its reference batch are therefore exchangeable draws from P_θ.

Following the simulation-based normalized-score construction [47, 10], define

$$E_i^{(\theta)} = \frac{(K+1)T_i^{(\theta)}(Y_i)}{T_i^{(\theta)}(Y_i) + \sum_{j=1}^K T_i^{(\theta)}(\widetilde{Y}_{ij}^{(\theta)})}, \tag{8}$$

$$M_0^{(\theta)} = 1, \qquad M_n^{(\theta)} = \prod_{i=1}^n E_i^{(\theta)}.$$

Each factor compares the observed score with the average score across the observation and its references. Conditional exchangeability makes this factor a conditional e-value, as established below.

Proposition 4.1 (Exact validity)**.** *For every $\theta \in \Theta$ and every sequence of positive predictable scores $(T_i^{(\theta)})_{i\geq 1}$, the process $(M_n^{(\theta)})_{n\geq 0}$ defined by (7) and (8) is a test martingale under H_θ.*

Proof. Fix θ and condition on $\mathcal{F}_{i-1}$ under H_θ. Set $Z_{i0} = Y_i$ and $Z_{ij} = \widetilde{Y}_{ij}^{(\theta)}$ for $j = 1, \ldots, K$. The batch is conditionally exchangeable, and $T_i^{(\theta)}$ is fixed given the past. Hence the normalized scores

$$W_{ij} = \frac{T_i^{(\theta)}(Z_{ij})}{\sum_{\ell=0}^{K} T_i^{(\theta)}(Z_{i\ell})}, \qquad j = 0, \ldots, K,$$

have equal conditional expectations and sum to one. Therefore,

$$\mathbb{E}_\theta[E_i^{(\theta)} \mid \mathcal{F}_{i-1}] = (K+1)\mathbb{E}_\theta[W_{i0} \mid \mathcal{F}_{i-1}] = 1.$$

The product construction in Section 3.1 gives the test-martingale property. □

Consequently, for every $\alpha \in (0, 1)$,

$$C_{n,\alpha} = \{\theta \in \Theta : M_n^{(\theta)} < \alpha^{-1}\} \tag{9}$$

satisfies the time-uniform coverage guarantee in (1). The reference simulations may share randomness across candidate parameters: the proof requires conditional exchangeability for each candidate under its own null, rather than independence across candidates.

Validity holds for every positive predictable score. The score determines expected log growth against false candidates, which we study next.

4.2 The optimal score

We assess a score through the expected log growth of its test martingale under P_{θ_*}. For a fixed positive measurable score T, define

$$\Gamma_K^{(\theta)}(T) = \mathbb{E}\left[\log \frac{(K+1)T(X_0)}{\sum_{j=0}^{K} T(X_j)}\right], \tag{10}$$

where $X_0 \sim P_{\theta_*}$ and $X_1, \ldots, X_K \overset{\text{i.i.d.}}{\sim} P_\theta$ are mutually independent. If the same score T is used at every step, then

$$\frac{1}{n}\mathbb{E}_{\theta_*}[\log M_n^{(\theta)}] = \Gamma_K^{(\theta)}(T).$$

Within the simulation-based normalized score construction (8), an *oracle score* maximizes this quantity for testing H_θ against P_{θ_*}.

The oracle characterization below specializes the log-optimality of normalized likelihood ratios for permutation invariance [24]. We give its classification proof to make the connection with InfoNCE explicit [45, 36]. The finite-K bound also follows by specializing the log-growth comparison of Dombowsky et al. [10].

Proposition 4.2 (Oracle score and finite-K gap)**.** *Assume $\mathrm{KL}(P_{\theta_*}\|P_\theta) < \infty$. For every positive measurable score T with $\Gamma_K^{(\theta)}(T) > -\infty$,*

$$\Gamma_K^{(\theta)}(T) \leq \Gamma_K^{(\theta)}(r_{\theta_*,\theta}) \leq \mathrm{KL}(P_{\theta_*}\|P_\theta).$$

The first inequality is an equality whenever $T = c\, r_{\theta_*,\theta}$ for some $c > 0$. Define

$$\Delta_K^{(\theta)} = \mathrm{KL}(P_{\theta_*}\|P_\theta) - \Gamma_K^{(\theta)}(r_{\theta_*,\theta}).$$

$\Delta_K^{(\theta)}$ measures the loss in expected log growth relative to the oracle likelihood-ratio martingale of Section 3.1 If

$$\chi^2(P_{\theta_*}\|P_\theta) := \int \{r_{\theta_*,\theta}(y) - 1\}^2 \, P_\theta(\mathrm{d}y) < \infty,$$

then

$$\begin{aligned} 0 \leq \Delta_K^{(\theta)} &\leq \log\left(1 + \frac{\chi^2(P_{\theta_*}\|P_\theta)}{K+1}\right) \\ &\leq \frac{\chi^2(P_{\theta_*}\|P_\theta)}{K+1}. \end{aligned} \tag{11}$$

In particular, under this moment condition, $\Gamma_K^{(\theta)}(r_{\theta_*,\theta}) \to \mathrm{KL}(P_{\theta_*}\|P_\theta)$ as $K \to \infty$.

Proof. Write $r = r_{\theta_*,\theta}$. Consider a classification problem with J uniform on $\{0, \ldots, K\}$. Conditional on J, draw X_J from P_{θ_*} and all remaining points independently from P_θ. For a score T, define

$$\pi_T(j \mid X_{0:K}) = \frac{T(X_j)}{\sum_{\ell=0}^{K} T(X_\ell)}.$$

By symmetry,

$$\Gamma_K^{(\theta)}(T) = \log(K+1) + \mathbb{E}[\log \pi_T(J \mid X_{0:K})].$$

Bayes' rule gives π_r as the true conditional distribution of J. Consequently,

$$\begin{aligned} &\Gamma_K^{(\theta)}(r) - \Gamma_K^{(\theta)}(T) \\ &\quad = \mathbb{E}[\mathrm{KL}(\pi_r(\cdot \mid X_{0:K})\|\pi_T(\cdot \mid X_{0:K}))] \geq 0. \end{aligned}$$

Multiplying r by a positive constant leaves π_r unchanged. The oracle objective is finite because the conditional entropy of J lies between zero and $\log(K+1)$.

For the gap, return to the sampling scheme in (10) and set

$$S_K = \frac{1}{K+1}\sum_{j=0}^{K} r(X_j).$$

Write $\mathbb{E}_{\text{null}}$ for expectation when all $K+1$ points are instead independent draws from P_θ. A change of measure and symmetry give

$$\begin{aligned} \Delta_K^{(\theta)} &= \mathbb{E}[\log S_K] \\ &= \mathbb{E}_{\text{null}}[r(X_0)\log S_K] \\ &= \mathbb{E}_{\text{null}}[S_K \log S_K] \geq 0. \end{aligned}$$

The final inequality follows from convexity and $\mathbb{E}_{\text{null}}[S_K] = 1$. If the chi-square divergence is finite, then under the original sampling scheme,

$$\mathbb{E}[S_K] = 1 + \frac{\chi^2(P_{\theta_*} \| P_\theta)}{K+1}.$$

Jensen's inequality gives $\Delta_K^{(\theta)} \le \log \mathbb{E}[S_K]$. The remaining bound follows from $\log(1+x) \le x$. □

The classification interpretation explains why the likelihood ratio is optimal for every K: it assigns the correct probability to each point being the draw from P_{θ_*}. The same principle underlies contrastive ratio estimation [11]. Moreover, $0 \le \Gamma_K^{(\theta)}(r_{\theta_*,\theta}) \le \log(K+1)$, so a finite reference batch limits the expected log evidence obtainable from each observation. Equation (11) quantifies the resulting gap.

4.3 Neural ratio scores

The oracle score depends on the unknown θ_* and on likelihoods that cannot be evaluated. We approximate it in two steps: replace θ_* by an estimate based on past information, and estimate the resulting likelihood ratio with a neural network. Specifically, define

$$T_i^{(\theta)}(y) = \exp\Big\{ s_{\phi_{i-1}}(\widehat{\theta}_{i-1}, y) - s_{\phi_{i-1}}(\theta, y) \Big\}, \quad (12)$$

where s_ϕ is the likelihood-to-evidence network of Section 3.2, and both $\widehat{\theta}_{i-1}$ and ϕ_{i-1} are $\mathcal{F}_{i-1}$-measurable.

At the population target (5), the term $-\log m(y)$ cancels in the score difference. Thus the target of (12) is $r_{\widehat{\theta}_{i-1},\theta}(y)$.

We train the likelihood-to-evidence network s_ϕ using the NRE-C objective [30] on independent simulations before monitoring, and then keep it fixed. At each step, we choose the alternative $\widehat{\theta}_{i-1}$ by minimizing the accumulated evidence over the evaluation grid. This choice is related to ME-estimation and e-posterior methods [50, 18]. Other predictable estimates and network updates are also permitted.

Table 1 summarizes the proposed conditional e-value and the likelihood-ratio based benchmarks used in our experiments.

4.4 Overall procedure

Algorithm 1 summarizes our implementation. We update the martingales over the entire grid, including previously excluded candidates, because the next alternative is selected by minimizing over this grid.

The output satisfies $\bar{C}_{n,\alpha} = \Theta_{\text{eval}} \cap \bigcap_{i=1}^n C_{i,\alpha}$. Consequently, by Theorem 4.1, $(\bar{C}_{n,\alpha})$ satisfies (1) for every $\theta_* \in \Theta_{\text{eval}}$. Coverage beyond the evaluation grid requires additional control between grid points; see Section C.2.2.

Table 1: Conditional e-values for candidate θ. Write $\widetilde{Y}_{i0}^{(\theta)} = Y_i$; the remaining K observations are fresh reference draws from P_θ. The neural score $T_i^{(\theta)}$ is defined in (12). Oracle LR and oracle score are the oracle variants of Direct and Conformal LR, respectively.

Construction	Conditional e-value $E_i^{(\theta)}$
Oracle LR (optimal)	$r_{\theta_*,\theta}(Y_i)$
Direct LR	$r_{\widehat{\theta}_{i-1},\theta}(Y_i)$
Oracle score	$\dfrac{(K+1) r_{\theta_*,\theta}(Y_i)}{\sum_{j=0}^K r_{\theta_*,\theta}(\widetilde{Y}_{ij}^{(\theta)})}$
Conformal LR	$\dfrac{(K+1) r_{\widehat{\theta}_{i-1},\theta}(Y_i)}{\sum_{j=0}^K r_{\widehat{\theta}_{i-1},\theta}(\widetilde{Y}_{ij}^{(\theta)})}$
Proposed (Conformal NRE)	$\dfrac{(K+1) T_i^{(\theta)}(Y_i)}{\sum_{j=0}^K T_i^{(\theta)}(\widetilde{Y}_{ij}^{(\theta)})}$

Section C discusses extensions to non-i.i.d. observations, nuisance parameters, and averaging over reference batches to reduce simulation noise.

5 EXPERIMENTS

We first use synthetic experiments to assess the efficiency costs of conformalization, learning likelihood ratios, and estimating the alternative parameter. Analytical or numerical likelihood evaluation makes these comparisons possible, with the true parameter providing an oracle alternative. We then turn to applications with intractable likelihoods, moving from a semi-synthetic audit of model substitution in an LLM API using real model replies, to stellar mass-function inference from real microlensing observations. Beyond assessing coverage where the truth is known, we examine how the confidence sequences narrow as observations accumulate, comparing neural and application-specific scores within the SLICE construction.

Throughout, we set $\alpha = 0.1$ and follow Algorithm 1, evaluating running-intersection confidence sets on finite candidate grids and using the minimum-evidence alternative by default. Learned likelihood-ratio estimators are fitted on independent data before monitoring and held fixed thereafter. Where the true parameter is known, we report cumulative miscoverage: the fraction of repetitions in which the method has excluded it by time n. We assess efficiency through confidence-set widths and candidate retention. Reported widths and endpoints are averaged over repetitions with nonempty sets. Implementation details,

Algorithm 1 Likelihood-free confidence sequence with neural ratio scores

Require: Simulator for P_θ; fixed finite grid Θ_{eval}; pretrained network s_ϕ; number of reference draws $K \geq 1$; level $\alpha \in (0,1)$.

Initialize $M_0^{(\theta)} \leftarrow 1$ for every $\theta \in \Theta_{\text{eval}}$, and $\bar{C}_{0,\alpha} \leftarrow \Theta_{\text{eval}}$.
for $i = 1, 2, \ldots$ **do**
  Using the previous martingale values, choose

$$\widehat{\theta}_{i-1} \in \arg\min_{\vartheta \in \Theta_{\text{eval}}} M_{i-1}^{(\vartheta)}.$$

  Before observing Y_i or drawing the current reference samples, fix the score functions

$$T_i^{(\theta)}(y) \leftarrow \exp\Big\{ s_\phi(\widehat{\theta}_{i-1}, y) - s_\phi(\theta, y) \Big\}, \ \theta \in \Theta_{\text{eval}}.$$

  Observe Y_i.
  for all $\theta \in \Theta_{\text{eval}}$ **do**
    Draw a fresh reference batch $\widetilde{Y}_{i1}^{(\theta)}, \ldots, \widetilde{Y}_{iK}^{(\theta)} \overset{\text{i.i.d.}}{\sim} P_\theta$.
    Compute the one-step e-value (8)
    Update $M_i^{(\theta)} \leftarrow M_{i-1}^{(\theta)} E_i^{(\theta)}$.
  end for
  Update the running intersection

$$\bar{C}_{i,\alpha} \leftarrow \bar{C}_{i-1,\alpha} \cap \Big\{ \theta \in \Theta_{\text{eval}} : M_i^{(\theta)} < \alpha^{-1} \Big\}.$$

  Output $\bar{C}_{i,\alpha}$.
end for

score diagnostics and additional experiments are provided in Sections D to H.

Synthetic experiments. We consider unit-variance Gaussian and likelihood-intractable g-and-k location models, each with a scalar parameter $\theta \in \Theta = [-4, 4]$, and SLCP, a five-parameter SBI benchmark with a complex, multimodal posterior on $\Theta = [-3, 3]^5$ [33]. For benchmarking, likelihoods are evaluated in closed form for Gaussian and SLCP, and by numerical inversion for g-and-k. We use an NRE-C score and compare against direct and conformalized likelihood-ratio confidence sequences, together with infeasible oracle variants using θ_* as the alternative. All three models use $n = 100$ observations over 1,000 repetitions and $K = 15$ reference draws per candidate per step.

For the two location models, we use 81 equally spaced candidates. Figure 1 shows results at $\theta_* = 0$; other true parameters and ablations are reported in Section F.2. Empirical cumulative miscoverage remains below α. Direct and conformalized LR confidence sequences have nearly identical hull widths and candidate retention, indicating little efficiency loss from conformalization. Replacing the likelihood ratio with the NRE-C score yields essentially the same results, while using the true parameter as the alternative gives narrower sets.

For SLCP, we use 5,120 scrambled Sobol candidates on $\Theta = [-3, 3]^5$, including $\theta_* = (0, 0, 2.2, 2.2, 0)$ and its three sign variants, which give the same distribution because θ_3 and θ_4 enter only through their squares. Because the fixed initial alternative is far from θ_* in this model, we use the first two observations only to choose the alternative and start the e-processes at $n = 3$ (Section E.3). Figure 2 shows cumulative miscoverage below α and similar rates of candidate exclusion for conformal NRE-C and conformal LR, with direct LR shrinking faster. Here conformalization has a visible cost, the price of obtaining validity from simulations rather than from the likelihood, whereas learning the ratio again costs little. The retained fractions level off as only a few candidates remain, and the gap to the oracle narrows. At $n = 100$, conformal NRE-C contains θ_* in 94.6% of repetitions, and keeps θ_* while dropping at least one of its three mirror images in only 1.2%, preserving the model's multiple modes (Section F.3).

Auditing model routing in an LLM API. We illustrate our method in a semi-synthetic audit of an API that may substitute a cheaper language model R for its advertised model P (16, 4; IRIS, 55). The auditor can sample from both models but cannot evaluate either likelihood. The aim is to estimate the substitution fraction θ while allowing the auditor to monitor incoming replies and stop adaptively. We take $P =$ `qwen3:8b`, one of IRIS's reference models, and $R =$ `qwen3:1.7b`, and use two of IRIS's prompts requesting random outputs whose model-specific reply distributions make substitution detectable. For each prompt, each query independently yields a reply from R with probability θ and P otherwise, giving the reply pmf $f_\theta = (1-\theta)p + \theta r$, where p and r are the models' respective reply pmfs. Likelihood ratios between candidate mixtures depend only on $\rho = r/p$, which we estimate from reply counts or a non-amortized binary NRE classifier distinguishing P from R, and use to construct the scores $T_i^{(\theta)}$ (Section G.3). We construct 200 streams of $n = 300$ observations per prompt at each $\theta_* \in \{0, 0.1, 0.3\}$ by mixing held-out replies from R and P with probabilities θ_* and $1 - \theta_*$. We compute confidence sequences using Algorithm 1, with $K = 10$ references per step from separate held-out reply pools. This semi-synthetic design preserves real model outputs while allowing coverage assessment at known substitution fractions. We compare with our implementation of IRIS's fixed-time confidence inter-

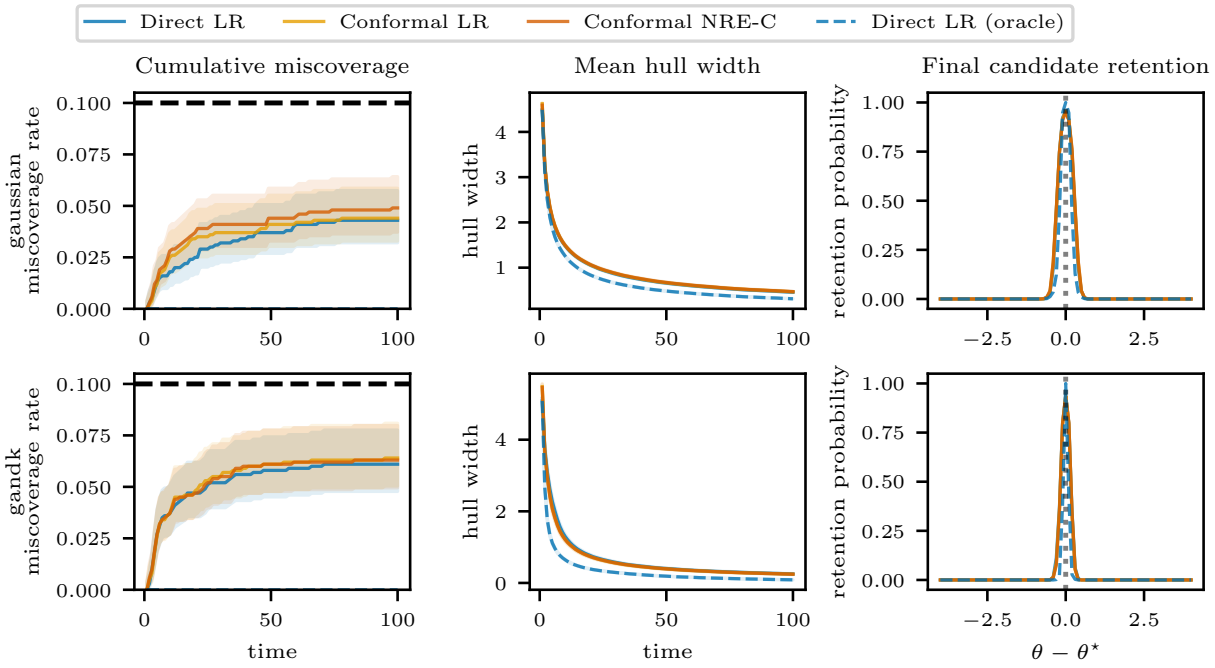

Figure 1: Confidence sequences for the Gaussian (top) and g-and-k (bottom) models at $\theta_* = 0$. Left: cumulative miscoverage (Wilson 95% bands; horizontal dashed line: $\alpha = 0.1$). Center: mean hull width over repetitions with nonempty sets. Right: final-time retention probability for each candidate. Solid curves: min-evidence alternative; dashed curves: oracle alternative.

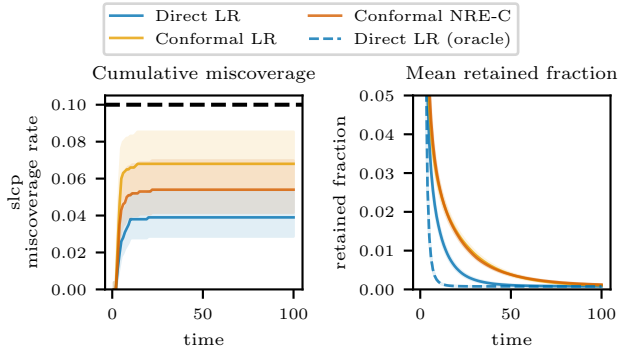

Figure 2: Confidence sequences for SLCP. Left: cumulative miscoverage (Wilson 95% bands; horizontal dashed line: $\alpha = 0.1$; oracle omitted). Right: mean fraction of candidates retained. Solid curves: min-evidence alternative; dashed curves: oracle alternative.

val (IRIS), which estimates θ from the share of replies our count-ratio estimate flags as atypical for P, and with an extension of IRIS's betting test to a confidence sequence for θ (Betting IRIS), which is only approximately valid because it uses estimated indicator rates (Section G.5). Figure 3 shows results at $\theta_* = 0.1$ for the digit prompt. The conformal sequences and Betting IRIS keep cumulative miscoverage below 0.1, whereas IRIS exceeds this level under continuous monitoring (panel a). The conformal count-ratio and

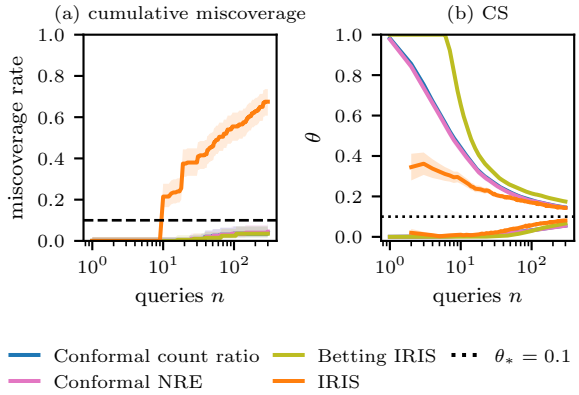

Figure 3: LLM routing audit at $\theta_* = 0.1$, 8 random digits prompt (200 repetitions). (a) Cumulative miscoverage (Wilson 95% bands; dashed $\alpha = 0.1$). (b) Mean lower and upper endpoints of the confidence sets over repetitions with nonempty sets, with 95% intervals for the means. Both prompts, and the IRIS conventions, are in Fig. 19.

NRE sequences have nearly identical CS hulls, both narrower than Betting IRIS (panel b). They also de-

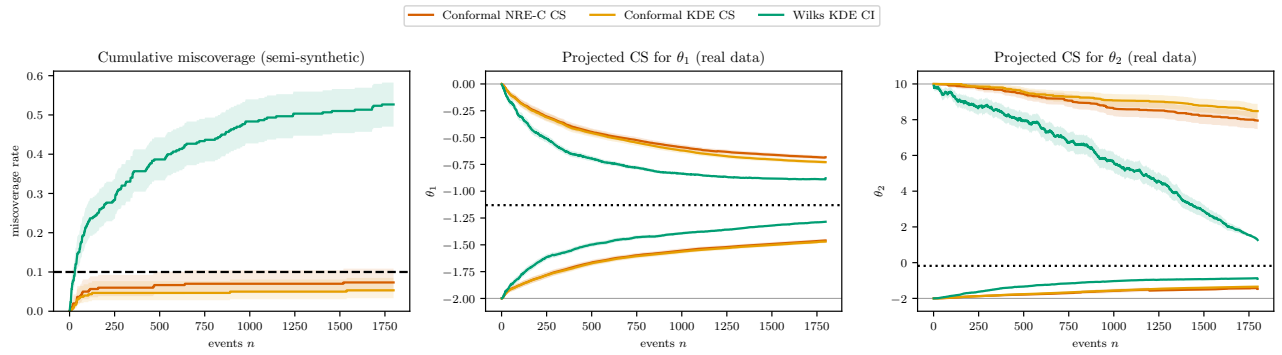

Figure 4: OGLE-IV microlensing. Left: cumulative miscoverage of the semi-synthetic runs (Wilson 95% bands; dashed $\alpha = 0.1$). Center and right: projected confidence sequences for θ_1 and θ_2 on the real data (mean endpoints over random orderings, with 95% intervals for the mean); grey lines mark the box and dotted lines the published $\theta_1 = -1.13$ and $\theta_2 = -0.18$ [26].

tect substitution sooner, as measured by the number of queries needed to exclude $\theta = 0$ (Table 7). The number prompt behaves alike (Fig. 19), and results at $\theta_* \in \{0, 0.3\}$ show consistent coverage for the conformal sequences and Betting IRIS, with faster detection also at 0.3, where Betting IRIS ends with slightly narrower sets (Figs. 18 and 20).

Inferring the initial mass function from microlensing events. Gravitational microlensing occurs when a foreground object, the lens, magnifies a background star, the source; the event's duration carries information about the lens mass, even when the lens is too faint to observe directly. We infer the slopes $\theta = (\theta_1, \theta_2)$ of the initial mass function of low-mass stars (θ_1) and brown dwarfs (θ_2) from the timescales of 1,798 OGLE-IV microlensing events toward the Galactic bulge [31], conditionally on each event's field and measurement error (Theorem C.1). The simulator is the Galactic model of Koshimoto et al. [26], as implemented in `genulens` [25]; its likelihood integrates over the distances, velocities, and masses of lenses and sources, and is intractable. A held-out half of the survey seasons is used to fit the model, which then passes a goodness-of-fit test on the analysed half (Section H.4), and simulation cost is reduced by reweighting per-field banks (Section H.3). The reweighting is based on the simulator's exact density ratio, and reweighted draws agree with direct simulation across the parameter range. We use $G = 2{,}048$ candidates on the box $[-2, 0] \times [-2, 10]$ and $K = 15$ references. Figure 4 compares the conformal NRE-C CS with two methods based on an accurate kernel density estimate (KDE) of the timescale's likelihood, possible because the timescale is one-dimensional: the conformal KDE CS, which uses the KDE likelihood ratio as its score, and the fixed-sample Wilks KDE interval, recomputed at every n. In 300 semi-synthetic repetitions, which simulate new timescales for the real events' covariates at the true value $\theta_* = (-1.1, 1.0)$ (left), both variants keep the empirical cumulative miscoverage below α, whereas the monitored Wilks interval soon exceeds it. On the real data, we show the projections of the sets (Section C.1.2) onto θ_1 (center) and θ_2 (right), averaged over 200 random orderings of the events. In both settings the NRE-C score is almost as efficient as the KDE score, and the Wilks interval is narrower, consistent with the additional cost of anytime-valid inference (Table 9). For θ_1, both sequences shrink steadily with n and contain the published value -1.13 [26] in 97% of orderings. For θ_2, they exclude low values but not high ones: a high θ_2 means few brown dwarfs, whose short events low-cadence fields rarely detect, so the data say little against it.

6 DISCUSSION

SLICE decouples validity from efficiency: normalizing any positive predictable score against fresh simulations yields exact likelihood-free confidence sequences, while the score only determines how fast they shrink. Neural ratio scores with a minimum-evidence alternative make a solid application-agnostic default: they came close to exact likelihood ratios on synthetic benchmarks and to application-specific scores in two applications with real data. Limitations include the reliance on a grid of candidates, the randomness that reference sampling adds to the sequences, and the need for a good predictable alternative to reach the best efficiency, and, as for any model-based method, a correctly specified simulator. Extensions addressing the first two are discussed in Section C, while addressing the third, for example by incorporating prior information, is a valuable direction for future work.

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Sequential Likelihood-Free Inference via Conformal E-Values: Supplementary Materials

A ADDITIONAL RELATED WORK

Simulation-based inference and posterior calibration. Classical likelihood-free methods include approximate Bayesian computation, which compares observed and simulated summaries, and synthetic likelihood, which models the distribution of summaries using repeated simulation [2, 53]. Neural simulation-based inference learns posterior, likelihood, or density-ratio surrogates, often amortizing inference across observations or parameter values [6]. Finite-budget posterior approximations can be overconfident or otherwise poorly calibrated [22]. Proposed remedies include balanced neural ratio estimation [9], differentiable coverage calibration [12], and conformalized posterior regions [34]. In particular, CANVI provides marginal coverage under the simulator-induced joint distribution of parameters and observations. This differs from coverage at each fixed data-generating parameter, and from coverage that holds simultaneously over observation times.

Density-ratio and contrastive estimation. Direct density-ratio estimation avoids separately estimating two densities [43]. Noise-contrastive estimation learns unnormalized models by discriminating data from noise [19, 20]; related classifier constructions estimate likelihood or posterior ratios from simulator output [5, 44, 21]. Contrastive formulations connect neural posterior and ratio estimation and provide multi-sample training objectives [11, 30]. Our classification interpretation of expected log growth is also connected to InfoNCE [45, 36]. These connections motivate efficient scores; validity follows from the subsequent normalization, independently of whether training recovers the population-optimal ratio.

Calibration of frequentist confidence regions. ACORE estimates odds ratios with probabilistic classifiers and calibrates tests through simulated critical values [7]. LF2I provides a modular framework for confidence-set construction and coverage diagnostics [8]. WALDO uses prediction algorithms or posterior estimators to construct a Wald-type statistic, followed by simulated quantile calibration [29]. For methods using learned critical values, coverage accuracy depends on the calibration step as well as the statistic. The TRUST framework distinguishes finite-sample local coverage from parameter-conditional coverage obtained asymptotically as the simulation budget increases, under appropriate assumptions [3]. These guarantees concern a fixed observed sample size. SLICE instead constructs conditional e-values at each observation and combines them into test martingales.

E-values and confidence sequences. Test martingales and Ville's inequality provide classical foundations for sequential inference [46]. Modern treatments develop confidence sequences, e-values, and guarantees under continuous monitoring and optional stopping [23, 48, 38]. Universal inference constructs finite-sample-valid likelihood-ratio-type tests and confidence sets through sample splitting, with extensions to sequential inference [51]. SLICE shares the use of nonnegative evidence processes and test inversion, obtaining the required conditional expectation property through simulator draws and exchangeability.

Exchangeability and neighboring sequential methods. Normalized scores have precedents in early transductive inference [15], conformal e-prediction and e-testing [47, 49], and testing under group invariance [24]. Sequential predictive two-sample tests also use learned classifiers to construct anytime-valid evidence [35]. SLICE applies sequential testing separately to candidate simulator distributions and inverts the tests for parameter inference. A different sequential setting is studied by Fischer and Ramdas [13], who develop anytime-valid Monte Carlo tests as additional resamples are generated for a fixed observed dataset. In SLICE, the observed data themselves arrive sequentially, with fresh reference simulations at each round, as in the simulation-based sequential construction of Dombowsky et al. [10].

B BACKGROUND

B.1 Coverage of the confidence sequence

The argument is standard [23, 38]; we include it for completeness. Fix $\theta_* \in \Theta$. By definition of $C_{n,\alpha}$,

$$\{\exists n \geq 1 : \theta_* \notin C_{n,\alpha}\} = \{\exists n \geq 1 : M_n^{(\theta_*)} \geq \alpha^{-1}\} \subseteq \{\sup_{n\geq 0} M_n^{(\theta_*)} \geq \alpha^{-1}\}. \tag{13}$$

Since $(M_n^{(\theta_*)})_{n\geq 0}$ is a test supermartingale under P_{θ_*}, Ville's inequality [46] gives

$$\mathbb{P}_{\theta_*}(\exists n \geq 1 : \theta_* \notin C_{n,\alpha}) \leq \mathbb{P}_{\theta_*}\left(\sup_{n\geq 0} M_n^{(\theta_*)} \geq \alpha^{-1}\right) \leq \alpha. \tag{14}$$

As θ_* was arbitrary, $\mathbb{P}_\theta(\forall n \geq 1 : \theta \in C_{n,\alpha}) \geq 1 - \alpha$ for every $\theta \in \Theta$, which is the claimed coverage.

B.2 Neural ratio estimation losses

We use q, m, and s_ϕ as defined in Section 3.2. Probabilities and expectations below refer to the specified auxiliary classification experiments. Population minimizers are characterized over unrestricted measurable scores; in practice, the score is restricted to the network family s_ϕ. For these characterizations, assume that $f_\theta(y)/m(y)$ is positive and finite for $q(\theta)m(y)\,\mathrm{d}\theta\,\mu(\mathrm{d}y)$-almost every (θ, y). Identities between population targets are understood almost everywhere under this measure.

Binary logistic loss (NRE-A). Introduce a label $C \in \{0, 1\}$ with $\mathbb{P}(C = 1) = \frac{1}{2}$, and let (θ, Y) have conditional density $q(\theta)f_\theta(y)$ given $C = 1$ and $q(\theta)m(y)$ given $C = 0$ [21]. A classifier with logit $s(\theta, Y)$ models the conditional probability of class 1 by $\sigma(s(\theta, Y))$, where $\sigma(u) = (1 + e^{-u})^{-1}$. Its population cross-entropy, dropping the overall factor $\frac{1}{2}$, is

$$\begin{aligned}\mathcal{L}_{\mathrm{bin}}(s) = &- \mathbb{E}_{\mathrm{joint}}[\log \sigma(s(\theta, Y))] \\ &- \mathbb{E}_{\mathrm{marg}}[\log\{1 - \sigma(s(\theta, Y))\}],\end{aligned} \tag{15}$$

where the expectations correspond to the joint density $q(\theta)f_\theta(y)$ and the product of marginals $q(\theta)m(y)$, respectively. Pointwise in (θ, y) on the common support, the integrand is minimized at

$$\sigma(s^\star(\theta, y)) = \frac{f_\theta(y)}{f_\theta(y) + m(y)}, \quad \text{i.e.} \quad s^\star(\theta, y) = \log \frac{f_\theta(y)}{m(y)}, \tag{16}$$

which agrees with (5). Under unequal sampling with $\mathbb{P}(C = 1) = \pi \in (0, 1)$, the same computation shows that the minimizer of the correspondingly weighted cross-entropy picks up the additive constant $\log\{\pi/(1 - \pi)\}$.

InfoNCE/NRE-B loss. An InfoNCE-type loss arises from a $(K + 1)$-way classification problem with one positive and K negatives [45, 36]. Fix θ_0, θ_1 such that P_{θ_0} and P_{θ_1} are mutually absolutely continuous. Draw $J \sim \mathrm{Unif}\{0, \ldots, K\}$; conditional on J, draw $X_J \sim P_{\theta_1}$ and $X_\ell \sim P_{\theta_0}$ independently for $\ell \neq J$. Bayes' rule gives

$$\mathbb{P}(J = j \mid X_{0:K}) = \frac{r_{\theta_1,\theta_0}(X_j)}{\sum_{\ell=0}^K r_{\theta_1,\theta_0}(X_\ell)}. \tag{17}$$

Consider a conditional class distribution parametrized by h,

$$q_h(j \mid X_{0:K}) = \frac{e^{h(X_j)}}{\sum_{\ell=0}^K e^{h(X_\ell)}}. \tag{18}$$

The associated population cross-entropy is

$$\mathcal{L}_{\mathrm{NCE},K}(h) = -\mathbb{E}[\log q_h(J \mid X_{0:K})]\,, \tag{19}$$

which is minimized by q_h equal to the true conditional class distribution. Matching the two softmax forms yields $h^\star(y) = \log r_{\theta_1,\theta_0}(y) + a$ for some constant a, almost everywhere under P_{θ_0}. This is the classification problem behind the growth criterion (10), with observations as the contrasted items.

NRE-B [11] instead contrasts parameters at a fixed observation. Writing $K_{\mathrm{tr}} \geq 2$ for the number of candidate parameters, draw $\vartheta_1, \ldots, \vartheta_{K_{\mathrm{tr}}} \overset{\text{i.i.d.}}{\sim} q$ and, independently, $J \sim \mathrm{Unif}\{1, \ldots, K_{\mathrm{tr}}\}$. Conditional on J and $\vartheta_{1:K_{\mathrm{tr}}}$, draw $Y \sim P_{\vartheta_J}$. Bayes' rule now gives

$$\mathbb{P}(J = j \mid \vartheta_{1:K_{\mathrm{tr}}}, Y) = \frac{f_{\vartheta_j}(Y)/m(Y)}{\sum_{\ell=1}^{K_{\mathrm{tr}}} f_{\vartheta_\ell}(Y)/m(Y)}. \tag{20}$$

The corresponding model is

$$q_s(j \mid \vartheta_{1:K_{\mathrm{tr}}}, Y) = \frac{e^{s(\vartheta_j, Y)}}{\sum_{\ell=1}^{K_{\mathrm{tr}}} e^{s(\vartheta_\ell, Y)}}, \tag{21}$$

with population loss

$$\mathcal{L}_{\mathrm{B}}(s) = -\mathbb{E}[\log q_s(J \mid \vartheta_{1:K_{\mathrm{tr}}}, Y)]\,. \tag{22}$$

The same argument shows that its minimizers are

$$s^\star(\theta, y) = \log \frac{f_\theta(y)}{m(y)} + a(y), \tag{23}$$

where a is an arbitrary measurable real-valued function: matching the two softmax forms at each fixed Y determines s only up to a per-observation shift, which enters numerator and denominator as a common factor. Score differences $s^\star(\theta_1, y) - s^\star(\theta_0, y)$ nonetheless recover $\log r_{\theta_1,\theta_0}(y)$, as the same function of y cancels in the difference.

Multiclass NRE-C loss. NRE-C [30] removes the indeterminacy in y by adding a class in which the observation is independent of all candidate parameters. Fix $K_{\mathrm{tr}} \geq 1$ and $\gamma > 0$. Draw $\vartheta_1, \ldots, \vartheta_{K_{\mathrm{tr}}}$ independently from q and, independently, draw a label $C \in \{0, \ldots, K_{\mathrm{tr}}\}$ with

$$\mathbb{P}(C = 0) = \frac{1}{1+\gamma}, \qquad \mathbb{P}(C = j) = \frac{\gamma}{K_{\mathrm{tr}}(1+\gamma)}, \quad j = 1, \ldots, K_{\mathrm{tr}}. \tag{24}$$

Conditional on the candidates and $C = j \geq 1$, draw $Y \sim P_{\vartheta_j}$. For $C = 0$, draw Y independently of the candidates with density m. Thus γ is the prior odds of a nonzero class versus class zero.

The model class probabilities are

$$\begin{aligned} \pi_{s,\gamma}(0 \mid \vartheta_{1:K_{\mathrm{tr}}}, y) &= \frac{K_{\mathrm{tr}}}{K_{\mathrm{tr}} + \gamma \sum_{\ell=1}^{K_{\mathrm{tr}}} e^{s(\vartheta_\ell, y)}}, \\ \pi_{s,\gamma}(j \mid \vartheta_{1:K_{\mathrm{tr}}}, y) &= \frac{\gamma e^{s(\vartheta_j, y)}}{K_{\mathrm{tr}} + \gamma \sum_{\ell=1}^{K_{\mathrm{tr}}} e^{s(\vartheta_\ell, y)}}, \qquad j = 1, \ldots, K_{\mathrm{tr}}, \end{aligned} \tag{25}$$

and the population loss is

$$\mathcal{L}_{\mathrm{C}}(s) = -\mathbb{E}[\log \pi_{s,\gamma}(C \mid \vartheta_{1:K_{\mathrm{tr}}}, Y)]\,. \tag{26}$$

Bayes' rule gives the true conditional class probabilities by replacing $e^{s(\vartheta_j, y)}$ with $f_{\vartheta_j}(y)/m(y)$. Matching the odds of each nonzero class against class zero therefore identifies $s^\star(\theta, y) = \log\{f_\theta(y)/m(y)\}$ uniquely, as in (5), without an additive ambiguity. NRE-A is recovered at $K_{\mathrm{tr}} = 1$ and $\gamma = 1$. As $\gamma \to \infty$, class zero disappears and the remaining class probabilities reduce to those of NRE-B.

C PRACTICAL VARIANTS AND EXTENSIONS

C.1 Variants used in our experiments

C.1.1 Beyond i.i.d. observations

The presentation assumes independent observations. Extensions are possible whenever, under H_θ, the observed next datum and the pseudo-observations are conditionally exchangeable given the past. For Markov or state-space simulators, this requires simulating pseudo-observations conditionally on the same observed history or latent-filtering information.

The simplest such case, used in Section H, is inference conditional on observed covariates. Suppose the pairs (X_i, Y_i) are i.i.d., and that under H_θ the conditional law of Y_i given $X_i = x$ is $P_\theta(\cdot \mid x)$, while the law of X_i is left unspecified. Let $\mathcal{G}_{i-1} = \sigma(\mathcal{F}_{i-1}, X_i)$ add the current covariate to the past.

Corollary C.1 (Conditional validity)**.** *Fix θ. At each i, let the score $T_i^{(\theta)}$ be positive and $\mathcal{G}_{i-1}$-measurable, and draw $\widetilde{Y}_{i1}^{(\theta)}, \dots, \widetilde{Y}_{iK}^{(\theta)}$ i.i.d. from $P_\theta(\cdot \mid X_i)$, independently of Y_i given $\mathcal{G}_{i-1}$. Then $(M_n^{(\theta)})_{n\geq 0}$ with $E_i^{(\theta)}$ as in (8) is a test martingale with respect to $(\mathcal{G}_n)$ under H_θ, whatever the distribution of the covariates. Consequently, (9) is a $(1-\alpha)$ confidence sequence.*

Proof. Under H_θ, conditional on $\mathcal{G}_{i-1}$, the variables $Y_i, \widetilde{Y}_{i1}^{(\theta)}, \dots, \widetilde{Y}_{iK}^{(\theta)}$ are i.i.d. from $P_\theta(\cdot \mid X_i)$ and $T_i^{(\theta)}$ is fixed. The proof of Theorem 4.1 then applies verbatim with $\mathcal{G}_{i-1}$ in place of $\mathcal{F}_{i-1}$, giving $\mathbb{E}_\theta[E_i^{(\theta)} \mid \mathcal{G}_{i-1}] = 1$; Ville's inequality gives the confidence sequence. □

Conditioning discards only the information about θ carried by the covariates themselves, and removes the need to model their distribution.

C.1.2 Nuisance parameters and projections

If the target is a lower-dimensional parameter $\psi = \psi(\theta)$, a valid projected confidence sequence is

$$C_{n,\alpha}^{\psi} = \{\psi(\theta) : \theta \in C_{n,\alpha}\}. \tag{27}$$

This projection can be conservative but preserves coverage. For testing a composite null $H_0 : \theta \in \Theta_0$, the simple-null e-processes do not automatically combine by averaging, since $M_n^{(\eta)}$ is generally not an e-process under P_θ when $\eta \neq \theta$. A generic valid but often conservative is to consider the process

$$M_n^{\mathrm{comp}}(\Theta_0) = \inf_{\eta \in \Theta_0} M_n^{(\eta)}. \tag{28}$$

This process need not be a test martingale. Nevertheless, for every $\theta \in \Theta_0$, $M_n^{\mathrm{comp}}(\Theta_0) \leq M_n^{(\theta)}$, so Ville's inequality gives

$$\begin{aligned} &\mathbb{P}_\theta\big(\exists n \geq 1 : M_n^{\mathrm{comp}}(\Theta_0) \geq \alpha^{-1}\big) \\ &\quad \leq \mathbb{P}_\theta\left(\sup_{n\geq 0} M_n^{(\theta)} \geq \alpha^{-1}\right) \leq \alpha. \end{aligned} \tag{29}$$

Thus rejecting when $M_n^{\mathrm{comp}}(\Theta_0)$ crosses α^{-1} gives an anytime-valid level-α test of the composite null.

C.2 Further extensions

C.2.1 Averaging over independent reference batches

To reduce variability due to the reference simulations, one may average conditional e-values obtained with independent reference batches. At each time i, generate $B \geq 1$ mutually independent reference batches, each containing K independent draws from P_θ, independently of $(\mathcal{F}_{i-1}, Y_i)$. Let $E_{i,b}^{(\theta)}$ denote the factor obtained from batch b using (8), with the same observation Y_i and predictable score $T_i^{(\theta)}$ for every batch. Define

$$\bar{E}_i^{(\theta)} = \frac{1}{B}\sum_{b=1}^{B} E_{i,b}^{(\theta)}, \qquad \bar{M}_n^{(\theta)} = \prod_{i=1}^{n} \bar{E}_i^{(\theta)}, \qquad \bar{M}_0^{(\theta)} = 1. \tag{30}$$

By Theorem 4.1 and linearity of conditional expectation,

$$\mathbb{E}_\theta\left[\bar{E}_i^{(\theta)} \mid \mathcal{F}_{i-1}\right] = 1. \tag{31}$$

Hence $(\bar{M}_n^{(\theta)})_{n\geq 0}$ is a test martingale under $\mathbb{P}_\theta$ and can be inverted to obtain a confidence sequence. The factors share Y_i but are independent conditional on $(\mathcal{F}_{i-1}, Y_i)$; averaging therefore divides their conditional variance by B, at the cost of BK reference draws per observation and candidate parameter.

C.2.2 Grid evaluation and continuous parameter spaces

The theoretical confidence sequence in (9) is defined over all $\theta \in \Theta$. Evaluating it on a finite grid gives the stated coverage guarantee only for true parameters on that grid. Continuous-parameter coverage can instead be obtained by computing a *certified outer approximation*: an interval is excluded only when every parameter it contains is certified to have crossed the rejection threshold. We describe this extension for a compact interval $\Theta \subset \mathbb{R}$.

A common random field over parameters. Suppose the simulator admits a jointly measurable representation $S(\theta, U)$, where the distribution of U does not depend on θ and $S(\theta, U) \sim P_\theta$. At each round i, after fixing the positive predictable score family $T_i^{(\theta)}$, generate independent base draws $U_{i1}, \ldots, U_{iK}$, independently of $(\mathcal{F}_{i-1}, Y_i)$, and set

$$\widetilde{Y}_{ij}^{(\theta)} = S(\theta, U_{ij}). \tag{32}$$

Assume that the score family is jointly measurable in (θ, y). These common random numbers define $M_n^{(\theta)}$ simultaneously over Θ, whether or not a particular parameter has been evaluated numerically. For each fixed θ, the conditional exchangeability argument of Theorem 4.1 remains unchanged.

Write

$$\ell_m(\theta) = \log M_m^{(\theta)}, \qquad h_\alpha = \log(1/\alpha). \tag{33}$$

The ideal continuous running intersection is

$$\mathcal{C}_{n,\alpha} = \bigcap_{m=1}^{n} C_{m,\alpha} = \left\{ \theta \in \Theta : \max_{1 \le m \le n} \ell_m(\theta) < h_\alpha \right\}. \tag{34}$$

Certified outer approximation. At round n, let $\mathcal{R}_n$ be a finite, measurably selected collection of intervals $I \subseteq \Theta$. An interval may be included in $\mathcal{R}_n$ only if, for some $m \le n$, a certified lower bound satisfies

$$h_\alpha \le \underline{\ell}_m(I) \le \inf_{\theta \in I} \ell_m(\theta). \tag{35}$$

Here, certification means that the bound holds throughout the selected interval for every realized history, or on a common probability-one event. Define

$$\widehat{C}_{n,\alpha} = \Theta \setminus \bigcup_{s=1}^{n} \bigcup_{I \in \mathcal{R}_s} I. \tag{36}$$

Proposition C.2 (Certified continuous-parameter inversion)**.** *Suppose every excluded interval satisfies* (35)*, and all selection rules are measurable. Then* $(\widehat{C}_{n,\alpha})_{n \ge 1}$ *is nested and satisfies*

$$\mathcal{C}_{n,\alpha} \subseteq \widehat{C}_{n,\alpha} \qquad \textit{for every } n \ge 1. \tag{37}$$

Consequently, for every $\theta_* \in \Theta$*,*

$$\mathbb{P}_{\theta_*}\Big(\theta_* \in \widehat{C}_{n,\alpha} \textit{ for all } n \ge 1\Big) \ge 1 - \alpha. \tag{38}$$

Proof. If θ belongs to an interval excluded by round n, then (35) implies $\ell_m(\theta) \ge h_\alpha$ for some $m \le n$. Thus $\theta \notin \mathcal{C}_{n,\alpha}$, establishing the containment. Nesting follows from (36), and the coverage guarantee follows from Ville's inequality applied to $M^{(\theta_*)}$. □

Intervals may be selected and subdivided adaptively after observing the current data. No multiplicity adjustment is needed for the number of intervals or refinement steps: the numerical procedure only enlarges the ideal confidence sequence. In particular, a computation may stop at any prescribed or data-dependent budget, provided all unresolved regions are retained.

Constructing interval bounds. Define

$$d_{ij}(\theta) = \log T_i^{(\theta)}(S(\theta, U_{ij})) - \log T_i^{(\theta)}(Y_i).$$

Dividing the numerator and denominator of (8) by the observed score gives

$$\begin{aligned} \ell_m(\theta) = m \log(K+1) \\ - \sum_{i=1}^{m} \log\left(1 + \sum_{j=1}^{K} e^{d_{ij}(\theta)}\right). \end{aligned} \tag{39}$$

Suppose verified calculations provide upper bounds

$$d_{ij}(\theta) \le \overline{d}_{ij}(I) \qquad \text{for all } \theta \in I. \tag{40}$$

Monotonicity of log-sum-exp yields the valid lower bound

$$\begin{aligned} \underline{\ell}_m(I) = m \log(K+1) \\ - \sum_{i=1}^{m} \log\left(1 + \sum_{j=1}^{K} e^{\overline{d}_{ij}(I)}\right). \end{aligned} \tag{41}$$

Hence I may be excluded whenever $\max_{m \le n} \underline{\ell}_m(I) \ge h_\alpha$. Lower bounds on d_{ij} analogously give upper bounds on ℓ_m. If these upper bounds remain below h_α for every $m \le n$, the entire interval is certified to belong to $\mathcal{C}_{n,\alpha}$.

The required bounds may be obtained by propagating interval enclosures through the simulator and score network. An alternative is a verified local Lipschitz bound. If $I = [c_I - r_I, c_I + r_I]$ and ℓ_m is $L_m(I)$-Lipschitz throughout I, then

$$\underline{\ell}_m(I) = \ell_m(c_I) - L_m(I) r_I \tag{42}$$

is valid. The constant must control the full theta-dependent calculation, including the reference simulations; gradients evaluated only at finitely many points do not establish such a bound.

Location simulators and ReLU scores. The one-dimensional experimental models in Section E.1 (Gaussian and g-and-k) have the form

$$S(\theta, U) = \theta + q(U). \tag{43}$$

A sharper certificate is available when the score network is composed of affine maps and ReLU activations, with fixed input standardization and frozen batch-normalization statistics during evaluation. Residual additions preserve this structure.

For the neural score in (12), write $q_{ij} = q(U_{ij})$. Conditional on the realized history and reference draws,

$$\begin{aligned} d_{ij}(\theta) = s_{\phi_{i-1}}(\widehat{\theta}_{i-1}, \theta + q_{ij}) \\ - s_{\phi_{i-1}}(\theta, \theta + q_{ij}) \\ - s_{\phi_{i-1}}(\widehat{\theta}_{i-1}, Y_i) + s_{\phi_{i-1}}(\theta, Y_i). \end{aligned}$$

Each nonconstant network input follows an affine line as θ varies. The network restricted to such a line is continuous and piecewise affine. Consequently, there is a finite partition of Θ into intervals on which every d_{ij}, for $i \le m$ and $j \le K$, is affine.

Consider one such interval $I = [a, b]$, and write $d_{ij}(\theta) = u_{ij}\theta + v_{ij}$ on I. Equation (39) becomes

$$\begin{aligned} \ell_m(\theta) = m \log(K+1) \\ - \sum_{i=1}^{m} \log\left(1 + \sum_{j=1}^{K} e^{u_{ij}\theta + v_{ij}}\right). \end{aligned} \tag{44}$$

Since log-sum-exp of affine functions is convex, ℓ_m is concave on I. Therefore,

$$\inf_{\theta\in[a,b]} \ell_m(\theta) = \min\{\ell_m(a), \ell_m(b)\}. \tag{45}$$

In particular, rejection of both endpoints at the same time m certifies rejection of the entire interval. More generally, the chord inequality

$$\begin{aligned} \ell_m((1-t)a+tb) \geq (1-t)\ell_m(a) + t\ell_m(b), \\ 0 \leq t \leq 1, \end{aligned} \tag{46}$$

provides a lower bound throughout the interval.

The partition must account for activation changes in all relevant historical network evaluations. Such partitions can be constructed by propagating affine representations through the layers and splitting intervals at zeros of ReLU preactivations; see Sotoudeh and Thakur [42]. Within each resulting interval, the fixed-time rejection region $\{\theta : \ell_m(\theta) \geq h_\alpha\}$ is an interval, possibly empty. Its boundaries can be enclosed using certified one-dimensional optimization and root finding. Unresolved boundary regions are retained.

Two qualifications are necessary. First, arbitrary grid intervals may contain activation breakpoints, so (45) cannot be applied without verifying the affine partition. Second, $\max_{m\leq n} \ell_m$ need not be concave: endpoint crossings at different times do not establish rejection throughout the interval. The certificate is applied separately at each time, with certified exclusions retained thereafter.

Adaptive evaluation and numerical rigor. A practical implementation can first apply coarse interval bounds, subdivide unresolved intervals, and exploit (45) where activation patterns are certified. Full activation-partition enumeration is therefore optional, and may be computationally expensive.

When a new parameter or subinterval is evaluated, its historical factors must use the original reference draws and the original score states, including the historical alternatives and network parameters. Refinement changes the numerical representation of the confidence set, not the underlying martingale. The next alternative may use all information available from the completed round; an exact minimizer over Θ is unnecessary for validity.

Formal numerical certification requires enclosures that account for rounding error in network evaluations, activation boundaries, and threshold comparisons. In particular, ordinary floating-point breakpoint calculations alone do not establish (35). With sound deterministic bounds, no additional statistical error allowance is required. The accuracy of the learned ratio remains irrelevant to coverage: the certificates concern the actual chosen score, rather than its unknown population target.

D IMPLEMENTATION AND COMPUTE

All experiments are implemented in `PyTorch` and `NumPy`. Randomness is controlled throughout by a single base seed expanded via NumPy `SeedSequence` spawning into independent per-component and per-repetition streams, so every training run, reference simulation, and repetition is individually reproducible. All experiments were run on an Apple Silicon M4 Pro with 24 GB of memory, using PyTorch's Metal (MPS) backend for GPU acceleration.

Cost. Let $G = |\Theta_{\text{eval}}|$ be the number of candidates and K the number of references per candidate. At each step the procedure draws K references for every candidate, so GK simulator calls, and evaluates the score $(K+1)G$ times; updating the wealth and choosing the minimum-evidence alternative costs $O(G)$. Over n steps this gives $O(nGK)$ simulator calls and score evaluations, while memory stays $O(GK)$, since only the current wealth of each candidate and the current step's references must be kept. The network is trained once before monitoring, a fixed cost that does not grow with n. The procedure needs no sample size fixed in advance: by Theorem 4.1 the confidence sequence is valid at any stopping time.

E EXPERIMENTAL DETAILS FOR THE SYNTHETIC EXPERIMENTS

E.1 Simulators

Gaussian. The Gaussian simulator returns $Y = \theta + Z$ with $Z \sim \mathcal{N}(0,1)$ and $\theta \in \Theta = [-4,4]$. Its likelihood is available in closed form, so it serves as a tractable benchmark where every oracle quantity can be computed

exactly.

g-and-k. The g-and-k simulator [39] returns $Y = Q_\theta(Z)$ with $Z \sim \mathcal{N}(0,1)$ and

$$Q_\theta(z) = \theta + b\left(1 + c\tanh(gz/2)\right) z \left(1 + z^2\right)^k,$$

where $(b, g, k, c) = (1, 2, 0.5, 0.8)$ as is standard. Sampling is a single standard-normal draw pushed through Q_θ, but Q_θ admits no closed-form inverse, so the density can only be evaluated by numerical root solving, making this a canonical likelihood-free benchmark.

SLCP. The SLCP (simple likelihood, complex posterior) simulator [33], as specified in `sbibm` [28], has $\theta \in \Theta = [-3, 3]^5$ and returns $Y = (Y^{(1)}, \ldots, Y^{(4)}) \in \mathbb{R}^8$ with $Y^{(j)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(m_\theta, S_\theta)$,

$$m_\theta = (\theta_1, \theta_2), \qquad S_\theta = \begin{pmatrix} s_1^2 + \epsilon & \rho s_1 s_2 \\ \rho s_1 s_2 & s_2^2 + \epsilon \end{pmatrix},$$

where $s_1 = \theta_3^2$, $s_2 = \theta_4^2$, $\rho = \tanh(\theta_5)$, and $\epsilon = 10^{-6}$ ensures positive definiteness. The likelihood is available in closed form, but since θ_3 and θ_4 enter only through their squares, $(\theta_1, \theta_2, \pm\theta_3, \pm\theta_4, \theta_5)$ define the same distribution and the posterior has four symmetric modes.

E.2 Training the neural ratio score

Objective and architecture. We train the likelihood-to-evidence network $s_\phi(\theta, y)$ of Section 3.2 with the NRE-C objective (Section B.2), with cardinality $K_{\text{tr}} = 20$ and $\gamma = 1$. The network is a residual multilayer perceptron with three residual blocks of width 128, ReLU activations, with batch normalization. Inputs (θ, y) are standardized using statistics estimated from an initial batch of 5×10^4 simulator draws.

Training data and optimization. Training is simulation-based and streaming: at every optimizer step we draw a fresh minibatch of 1024 pairs $\theta \sim q$, $Y \sim P_\theta$, with q uniform on Θ, for a total of 1.024×10^7 simulated pairs per model. We use Adam with learning rate 5×10^{-4} and gradient-norm clipping at 5. The network is trained once per model before monitoring begins and kept fixed throughout; updating ϕ during monitoring is permitted provided every update at time i uses only $\mathcal{F}_{i-1}$, and affects efficiency but not validity.

Monitoring training quality via expected log growth. The natural figure of merit for a score used in the test martingale is its expected one-step log growth (10). For a score $T(y) = \exp\{h(y)\}$ with $X_0 \sim P_\vartheta$ and $X_1, \ldots, X_K \overset{\text{i.i.d.}}{\sim} P_\theta$, define

$$\mathcal{L}_K(h; \theta, \vartheta) = -\mathbb{E}\left[\log \frac{\exp\{h(X_0)\}}{\sum_{j=0}^K \exp\{h(X_j)\}}\right], \tag{47}$$

so that

$$\Gamma_K(T; \theta, \vartheta) = \log(K+1) - \mathcal{L}_K(h; \theta, \vartheta), \tag{48}$$

where $\Gamma_K(T; \theta, \vartheta)$ denotes (10) with P_{θ_*} replaced by P_ϑ. We therefore monitor training on a fixed held-out bank of 4,096 tuples $(\theta, \vartheta, X_{0:K})$ with $K = 15$, sampled half with independent prior draws (θ, ϑ) and half with local perturbations $\vartheta = \theta + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.25^2)$ clipped to Θ, and report the estimated negative expected log growth $-\Gamma_K = \mathcal{L}_K - \log(K+1)$ of the trained score against that of the exact (Gaussian) or numerical (g-and-k) likelihood ratio on the same bank. Since (48) shows that minimizing $\mathcal{L}_K$ maximizes the same finite-K growth used by the test martingale, $\mathcal{L}_K$ could also serve directly as a training objective; we do not pursue this here.

E.3 Running the confidence sequence

For the Gaussian and g-and-k models, the martingale is evaluated over a fixed grid Θ_{eval} of 81 equispaced candidates on $\Theta = [-4, 4]$ (see Section C for coverage at off-grid values). For SLCP, Θ_{eval} consists of 5,120 points of a scrambled Sobol sequence [41] on $\Theta = [-3, 3]^5$, in which $\theta_* = (0, 0, 2.2, 2.2, 0)$ and its three mirror images replace their nearest points; together these are the four parameters $(0, 0, \pm 2.2, \pm 2.2, 0)$, which define the same distribution because θ_3 and θ_4 enter the model only through their squares (Section E.1). In all models

we monitor $n = 100$ observations over 1,000 repetitions at level $\alpha = 0.1$ with $K = 15$ null simulations per candidate per step. Throughout, we report the nested (running-intersection) version of the confidence sequence, $C_{i,\alpha} \leftarrow C_{i-1,\alpha} \cap \{\theta \in \Theta_{\text{eval}} : M_i^{(\theta)} < \alpha^{-1}\}$: a candidate whose martingale has ever crossed α^{-1} stays excluded. Since Ville's inequality bounds the probability that $M^{(\theta_*)}$ *ever* exceeds α^{-1}, this intersection preserves anytime-validity while making the sets monotone.

Predictable alternative. Any estimator based only on past information can be used for $\widehat{\theta}_{i-1}$. For example, it may be a point estimator based on $Y_{1:i-1}$, a simulator-based posterior summary, or the minimum-evidence estimator

$$\widehat{\theta}_{i-1} \in \operatorname*{arg\,min}_{\vartheta \in \Theta_{\text{eval}}} M_{i-1}^{(\vartheta)}, \tag{49}$$

related to ME-estimation and e-posterior methods [50, 18]. These choices affect only efficiency, not validity. We use (49); Section F.2 compares it with model-based running point estimates.

Discarding the first observations (SLCP). The predictable alternative must be initialized before any data are observed, and we use a fixed default, $\widehat{\theta}_0 = (0, 0, 1, 1, 0)$ for SLCP. In this problem the default is far from θ_*, so the first one-step e-values sharply decrease the wealth of many candidates, which then takes many observations to recover. For SLCP we therefore discard the first two e-values and build the sets from $M_i^{(\theta)}/M_2^{(\theta)}$ for $i \geq 3$; the first two observations are still used to choose $\widehat{\theta}_2$, so the guarantee holds from $i = 3$ onwards. More generally, discarding a few initial e-values reduces the dependence on the starting value at the cost of a few observations.

Reference simulations. At each step, K fresh pseudo-observations are generated for every candidate, from random numbers shared across candidates.

Common random numbers. Draws are coupled across candidates by common random numbers: if the simulator is written as $Y = S(\theta, U)$ with U a vector of standard normals, we use the same U_{ij} across candidates and set $\widetilde{Y}_{ij}^{(\theta)} = S(\theta, U_{ij})$ (a single batch of standard normals per step, shifted by θ for the Gaussian, pushed through Q_θ for g-and-k, or mapped to $m_\theta + L_\theta z$ for SLCP, with L_θ the Cholesky factor of S_θ). This smooths the random field $\theta \mapsto M_n(\theta)$ over the grid without affecting coverage: for the true θ_* the pseudo-observations still have distribution P_{θ_*} and are independent across time, so the coverage proof is unaffected, and dependence of the auxiliary simulations across candidates does not compromise pointwise confidence-sequence validity.

E.4 Confidence-sequence diagnostics

Cumulative miscoverage. Cumulative miscoverage at time n is the fraction of repetitions in which the confidence sequence has excluded the true parameter θ_* at some time $i \leq n$; it is therefore nondecreasing in n, and anytime-validity requires it to stay below α uniformly. Figures show it with pointwise Wilson 95% intervals; the table columns headed "mis." report it at the final time and should be read against the nominal $\alpha = 0.1$.

Mean hull width. For the Gaussian and g-and-k models, hull width measures how tightly the procedure localises θ_*. At each time the confidence set is the collection of grid candidates not yet rejected, and we record the width of its convex hull, i.e. the largest retained candidate minus the smallest, averaged over the repetitions with a nonempty set. Figures trace it over time; tables report it early, at $n = 10$ (W_{10}), and at the final time (W_{100}).

Final candidate retention. For the Gaussian and g-and-k models, retention of a grid candidate θ is the fraction of repetitions in which θ remains in the confidence set at the final time, plotted against $\theta - \theta_*$; concentration around zero indicates that false candidates are excluded reliably.

Mean retained fraction. For SLCP, whose confidence set is a subset of a five-dimensional candidate set and so has no single width, we instead report the fraction of the candidates in Θ_{eval} that remain in the confidence set, averaged over all repetitions.

F ADDITIONAL RESULTS FOR THE SYNTHETIC EXPERIMENTS

F.1 Neural ratio quality (Gaussian and g-and-k)

For the Gaussian and g-and-k models, we assess the baseline network of Section E.2, separately for the contrastive (NRE-C) and binary (NRE-A) objectives. The score enters the martingale only through the log-ratio $s_\phi(\theta_1, y) - s_\phi(\theta_0, y)$, so we measure how well this recovers the true log-ratio and the log-growth it delivers. On all ordered pairs (θ_0, θ_1) from a 21-point equispaced subgrid of Θ (every fourth candidate of Θ_{eval}) we report three static, pair-level measures. *Excess BCE* is the network's binary cross-entropy on discriminating f_{θ_1} from f_{θ_0}, minus that of the optimal classifier (whose logit is the true log-ratio); this difference is zero exactly when the score recovers the log-ratio, and positive otherwise. *Log-ratio RMSE* and *bias* are the root-mean-square and mean signed error of $s_\phi(\theta_1, \cdot) - s_\phi(\theta_0, \cdot)$ against the exact (Gaussian) or numerical (g-and-k) log-ratio. We complement these with the dynamic *log-growth validation* of Section E.2: along training we track the negative expected one-step log-growth $-\Gamma_K$ of the current score against that of the oracle likelihood ratio on the held-out bank, reported on three subsets: *broad* pairs, with θ and ϑ drawn independently from the prior; *local* pairs, with $\vartheta = \theta + \varepsilon$ a small perturbation of θ; and *overall*, the equal (50/50) mixture of the two that makes up the full bank. A score that recovers the oracle drives every panel down to the oracle line.

These diagnostics explain why the neural confidence sequences in Figs. 1 and 13 are essentially indistinguishable from the exact-likelihood-ratio sequences: at baseline capacity both scores achieve near-oracle log-growth, even where they do not recover the log-ratio itself.

Pair-level accuracy. Excess BCE is small everywhere for both scores and both models, confirming the ratio classifiers have converged. The log-ratio RMSE and bias are near zero on and around the diagonal $\theta_0 \approx \theta_1$ and grow toward the off-diagonal corners, where NRE-C is generally the weaker of the two: its off-diagonal error exceeds NRE-A's on g-and-k in both RMSE and bias (Fig. 6) and on the Gaussian in bias (Fig. 5). These off-diagonal errors do not, however, reflect in the confidence sequences, which are near-identical across scores (Fig. 13). This is because efficiency is governed not by log-ratio accuracy but by the expected log-growth, and for well-separated laws the growth saturates: once a candidate is clearly distinct from the truth, any roughly correct score already achieves near-maximal growth, so the large off-diagonal errors leave efficiency essentially untouched. Pair-level accuracy is therefore the wrong lens for efficiency; the relevant object is the log-growth itself, which we examine next.

Log-growth validation. Tracking expected one-step log-growth along training (Figs. 7a–8b), both scores descend onto the oracle likelihood-ratio line in all three subsets (broad, local, and overall) for both models. Since expected log-growth is exactly the efficiency criterion of Proposition 4.2, this convergence directly explains why the neural sequences in Fig. 13 match the exact-likelihood-ratio sequences in width and retention.

F.1.1 Small architecture

We repeat these diagnostics for the small architecture of the ablations (Section F.2), a single hidden layer of two units.

Pair-level accuracy. Under the deliberately underparameterized network the two objectives separate sharply, reversing their baseline ordering. NRE-C's excess BCE stays near zero and its bias small on both models; its log-ratio RMSE stays low on g-and-k (Fig. 10), but on the Gaussian it grows in the far-apart corners about as much as NRE-A's (Fig. 9). NRE-A degrades markedly on both: its excess BCE is now visibly elevated, and its log-ratio RMSE and bias grow several-fold. The degradation is far worse on g-and-k, where NRE-A's RMSE is large across almost the entire grid. Crucially, this error is no longer confined to the far-apart, saturated corners that were harmless at baseline: on g-and-k it reaches the near-diagonal band, which drives localization and where growth is *not* saturated. This is what makes the small-architecture failure of NRE-A costly rather than cosmetic, as the log-growth confirms next.

Log-growth validation. The one-step log-growth (Figs. 11a–12b) makes the separation explicit. NRE-C moves toward the oracle likelihood-ratio line in all three subsets on both models: it reaches it on g-and-k, while on the Gaussian it is still approaching it when training stops, most visibly on the local pairs. NRE-A does

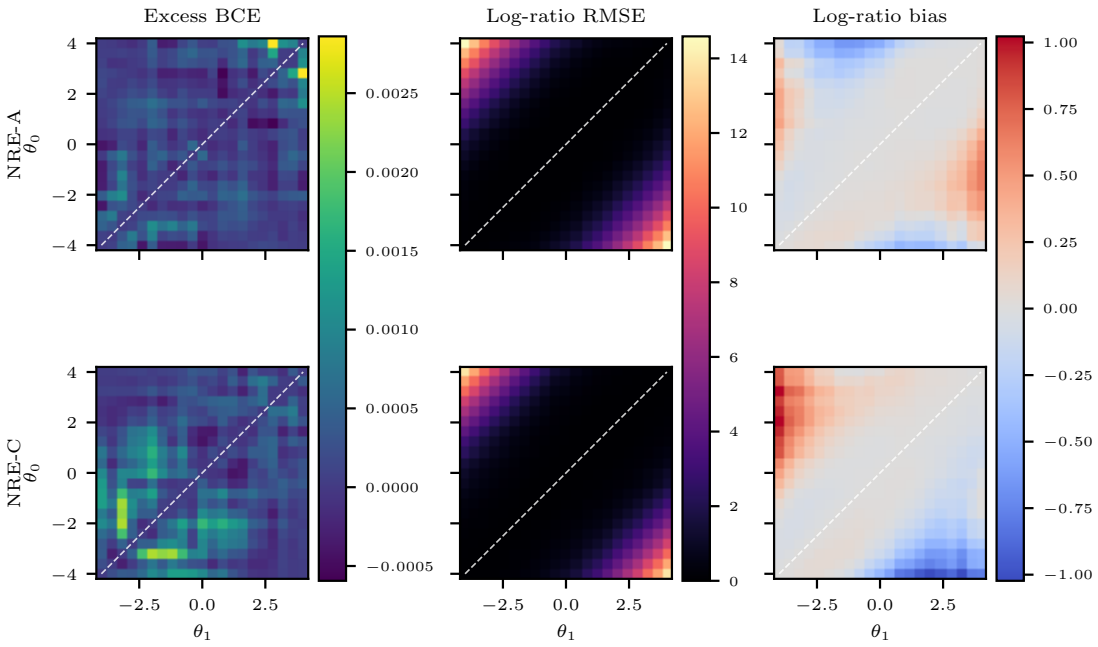


Figure 5: Gaussian model, baseline network. Per parameter-pair (θ_0, θ_1): excess BCE (left), log-ratio RMSE (center), and log-ratio bias (right), for NRE-A (top) and NRE-C (bottom).

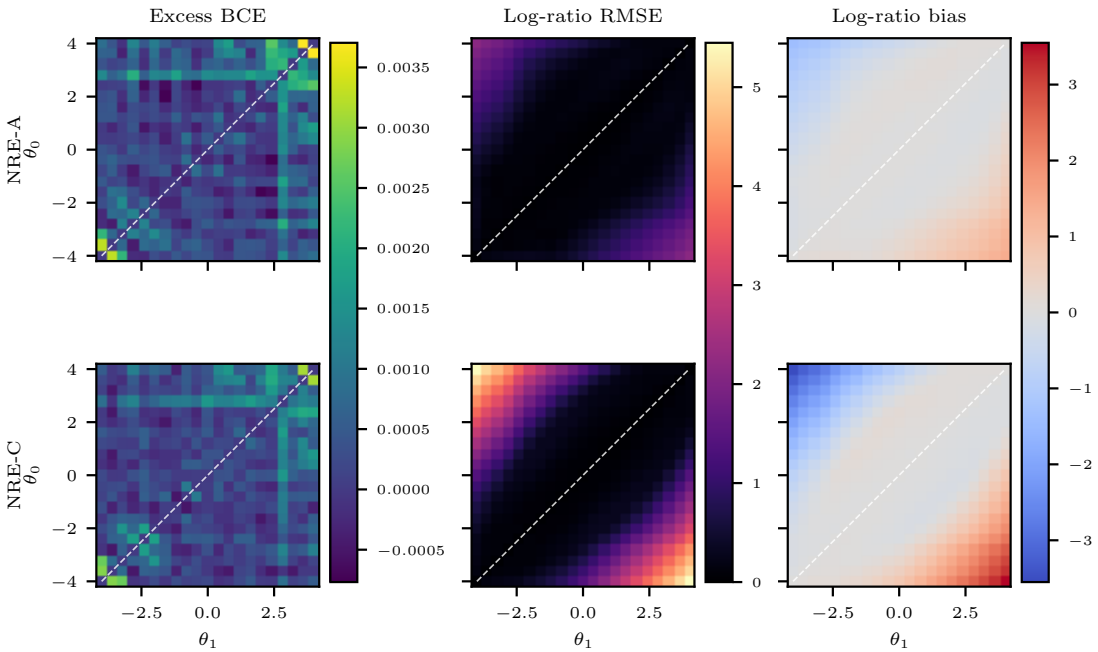
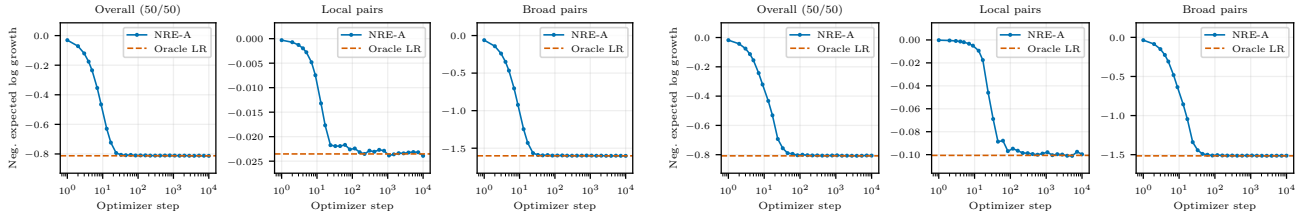
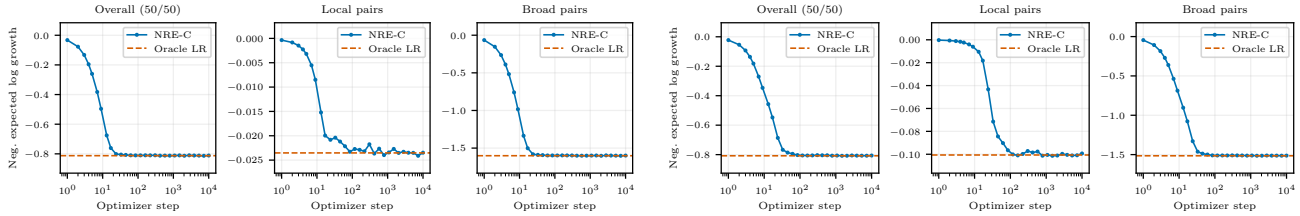


Figure 6: As Fig. 5, for the g-and-k model.



(a) Gaussian (b) g-and-k

Figure 7: NRE-A log-growth validation along training, baseline network. Negative expected one-step log-growth of the trained score (blue) against the oracle likelihood ratio (dashed), on broad, local, and overall pairs.



(a) Gaussian (b) g-and-k

Figure 8: As Fig. 7, for NRE-C.

not. On the Gaussian it plateaus visibly above the oracle in every subset, a persistent but bounded log-growth deficit. On g-and-k it collapses, remaining almost flat and capturing a small fraction of the available growth (its overall curve settles near -0.11 against an oracle near -0.81). Because this shortfall reaches the contested near-diagonal region rather than the saturated tails, it translates directly into wider, less concentrated confidence sequences under NRE-A. This is severe on g-and-k, where its sets fill much of the grid and barely localize θ_* (Table 2). Coverage nonetheless remains valid throughout: a misspecified score forfeits power, never validity, and the contrastive objective is markedly more robust to the misspecification.

F.2 Ablations (Gaussian and g-and-k)

F.2.1 Set-up

Because coverage is guaranteed pointwise at the true parameter, we repeat every ablation at three truths $\theta_* \in \{-2, 0, 2\}$, one central and two off-center in $\Theta = [-4, 4]$, to check that behavior does not depend on where the truth sits. Over 1,000 repetitions each, we vary one component of the baseline procedure at a time (minimum-evidence alternative, $K = 15$, the score network of Section E.2):

Score. The score $T_i^{(\theta)}$ determines efficiency but not validity, so we compare four choices of increasing practicality. The *direct exact-LR* martingale uses the true likelihood ratio without conformalization; its *conformalized* counterpart applies the soft-rank normalization of (8) to the same exact ratio, isolating the cost of conformalization alone. The two neural scores replace the likelihood ratio with a learned estimate: NRE-C, the contrastive estimator of the main text, and NRE-A, the binary-classification ratio estimator (Hermans et al. 21) (see Section B.2). Comparing all four separates the price of conformalization from the price of learning the ratio.

K. K is the number of fresh null simulations drawn per candidate at each step, forming the exchangeable reference batch against which the observation is scored. For the oracle score, the finite-K gap in expected log-growth vanishes as $K \to \infty$ (Theorem 4.2). We contrast the baseline $K = 15$ with the minimal $K = 1$ to examine how strongly the simulation budget affects the sequence.

$\widehat{\theta}_{i-1}$. The predictable alternative $\widehat{\theta}_{i-1}$ is the reference parameter against which each candidate θ is scored, and must depend only on $\mathcal{F}_{i-1}$. The baseline chooses it by minimum evidence, $\widehat{\theta}_{i-1} \in \arg\min_\vartheta M_{i-1}^{(\vartheta)}$, the candidate the accumulated evidence currently favors. We compare this against a simpler running point estimate of the

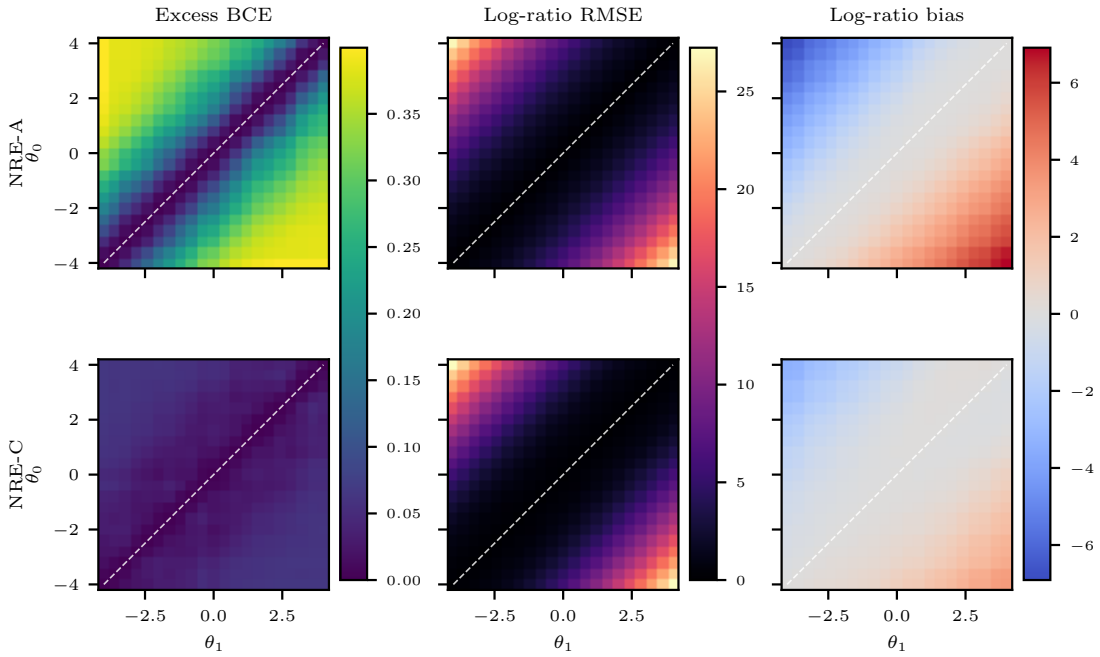


Figure 9: Gaussian model, small (underparameterized) network. Per parameter-pair (θ_0, θ_1): excess BCE (left), log-ratio RMSE (center), and log-ratio bias (right), for NRE-A (top) and NRE-C (bottom).

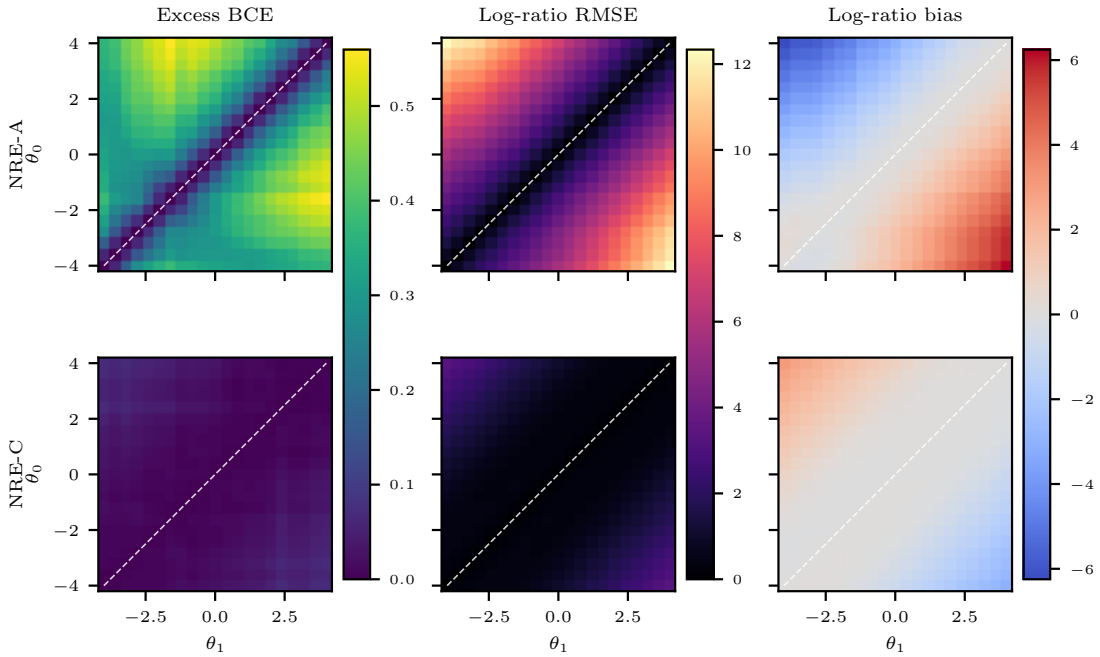
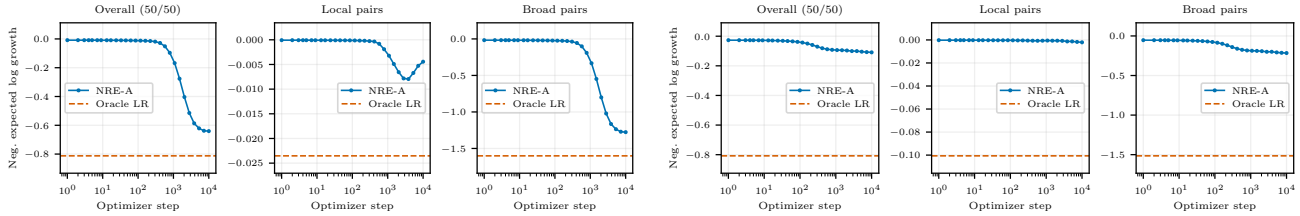
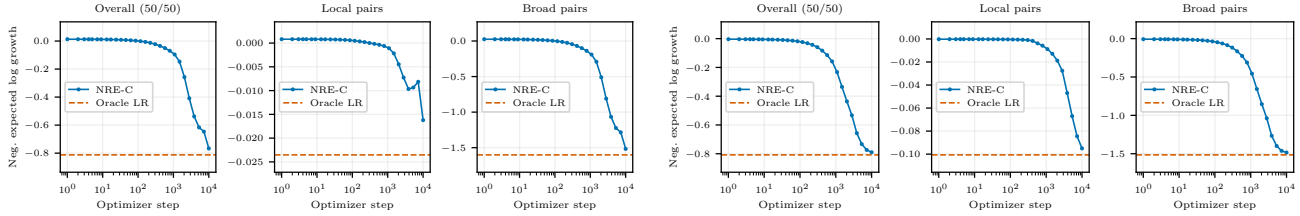


Figure 10: As Fig. 9, for the g-and-k model.



(a) Gaussian (b) g-and-k

Figure 11: NRE-A log-growth validation along training, small (underparameterized) network. Negative expected one-step log-growth of the trained score (blue) against the oracle likelihood ratio (dashed), on broad, local, and overall pairs.



(a) Gaussian (b) g-and-k

Figure 12: As Fig. 11, for NRE-C.

observations so far that does not require evaluating the martingale over the grid: the running mean for the Gaussian model and the running median for g-and-k.

Architecture. The neural scores are trained networks, so their quality depends on capacity. To probe robustness to a poorly estimated ratio, we replace the baseline network with a deliberately underparameterized "small" architecture i.e., a single hidden layer of two units, far too small to represent the true log-ratio, and ask whether coverage survives when the score is badly misspecified.

F.2.2 Aggregated results

Table 2 reports, for every ablation block at all three truths, cumulative miscoverage at $n = 100$ and the mean hull width at an early and the final time, W_{10} and W_{100}. Since the widths are conditioned on nonempty sets, Table 3 gives the corresponding rates: they never fall below 96%, so the conditioning is a negligible correction almost everywhere. The one exception is the underparameterized NRE-A on g-and-k at $\theta_* = 0$ and 2, whose sets never empty (100%) because it rejects almost nothing: with $W_{100} = 5.5$ on a width-8 grid, it retains most candidates and has essentially failed to localize θ_*.

Across the baseline block, the three conformal estimators agree to within Monte Carlo error in both miscoverage and width, at all three truths and on both models: neither neural score is distinguishable from the conformalized exact likelihood ratio. The direct exact-LR martingale departs from them in some cases, mainly at $\theta_* = \pm 2$, where it is more conservative and its sets are wider. Baseline miscoverage stays below 0.07, comfortably below the nominal 0.1, the conservativeness expected of Ville's inequality. Replacing minimum evidence with a model-based running estimate leaves the Gaussian results essentially unchanged, but widens the g-and-k sets substantially.

Reducing the null budget to $K = 1$ preserves validity but widens the sets, by roughly 20–40% at $n = 100$ on both models, while miscoverage rises toward the nominal level. Both follow from the added Monte Carlo noise in the single-draw e-value, false candidates are excluded more slowly, widening the sets, while the truth's own martingale fluctuates more and approaches the Ville bound. Neither effect breaches validity.

The small-architecture block is the clearest separation of validity from efficiency. Deliberately underparameterizing the score leaves NRE-C close to its well-specified baseline but degrades NRE-A, and the degradation is model-dependent: mild on the Gaussian (W_{100} up to 1.04 against a baseline near 0.56), but a collapse on g-and-k at $\theta_* = 0$ and 2, where NRE-A's miscoverage falls to essentially zero and W_{100} inflates to 5.5 on a grid

Table 2: Ablations at $\alpha = 0.1$ over 1,000 repetitions, $n = 100$ monitored observations. Each ablation block lists only the estimators it affects. Columns report cumulative miscoverage at $n = 100$, and mean hull width at $n = 10$ and $n = 100$.

	$\theta_* = -2$			$\theta_* = 0$			$\theta_* = 2$		
Estimator	mis.	W_{10}	W_{100}	mis.	W_{10}	W_{100}	mis.	W_{10}	W_{100}
Gaussian									
Baseline (min.-evidence, $K = 15$)									
Direct LR	0.049	1.885	0.584	0.043	1.486	0.458	0.034	1.922	0.593
Conformal LR	0.054	1.757	0.556	0.044	1.493	0.467	0.050	1.797	0.568
Conformal NRE-A	0.051	1.755	0.556	0.053	1.491	0.464	0.051	1.796	0.562
Conformal NRE-C	0.055	1.773	0.574	0.049	1.496	0.470	0.052	1.799	0.571
$K = 1$ null simulation									
Conformal LR	0.060	2.407	0.718	0.065	2.055	0.612	0.076	2.392	0.710
Conformal NRE-A	0.062	2.398	0.717	0.066	2.055	0.612	0.078	2.395	0.710
Conformal NRE-C	0.062	2.410	0.733	0.068	2.044	0.614	0.076	2.388	0.713
Model-based plug-in alternative									
Direct LR	0.048	1.886	0.581	0.045	1.484	0.454	0.033	1.922	0.594
Conformal LR	0.051	1.768	0.557	0.055	1.492	0.458	0.054	1.807	0.569
Conformal NRE-A	0.051	1.767	0.558	0.056	1.492	0.457	0.052	1.805	0.568
Conformal NRE-C	0.049	1.790	0.576	0.055	1.492	0.462	0.051	1.808	0.575
Small architecture									
Conformal NRE-A	0.078	2.161	0.752	0.043	2.229	0.843	0.064	2.470	1.038
Conformal NRE-C	0.058	1.782	0.637	0.058	1.625	0.584	0.055	1.872	0.677
g-and-k									
Baseline (min.-evidence, $K = 15$)									
Direct LR	0.040	1.968	0.384	0.061	1.303	0.252	0.027	1.400	0.284
Conformal LR	0.045	1.949	0.380	0.064	1.219	0.245	0.066	1.180	0.240
Conformal NRE-A	0.047	1.960	0.384	0.061	1.223	0.248	0.067	1.185	0.249
Conformal NRE-C	0.045	1.967	0.387	0.063	1.212	0.244	0.067	1.189	0.247
$K = 1$ null simulation									
Conformal LR	0.058	2.523	0.471	0.065	1.910	0.333	0.075	2.380	0.335
Conformal NRE-A	0.054	2.439	0.460	0.071	1.830	0.324	0.074	2.510	0.327
Conformal NRE-C	0.056	2.419	0.460	0.071	1.835	0.322	0.077	2.460	0.329
Model-based plug-in alternative									
Direct LR	0.042	2.300	0.564	0.041	1.825	0.456	0.023	1.932	0.484
Conformal LR	0.045	2.348	0.558	0.056	1.775	0.451	0.059	1.747	0.450
Conformal NRE-A	0.048	2.347	0.558	0.056	1.766	0.455	0.061	1.750	0.462
Conformal NRE-C	0.049	2.334	0.554	0.056	1.736	0.449	0.060	1.764	0.458
Small architecture									
Conformal NRE-A	0.001	6.739	1.067	0.000	6.157	5.501	0.000	6.147	5.500
Conformal NRE-C	0.052	2.001	0.389	0.055	1.290	0.248	0.055	1.314	0.250

Table 3: Percentage of nonempty confidence sets at $n = 100$, over 1,000 repetitions. Mean hull widths in Table 2 are averaged over these repetitions only. Rows follow the same blocks as Table 2.

	Gaussian			g-and-k		
Estimator	$\theta_* = -2$	$\theta_* = 0$	$\theta_* = 2$	$\theta_* = -2$	$\theta_* = 0$	$\theta_* = 2$
Baseline (min.-evidence, $K = 15$)						
Direct LR	98.1	99.9	98.8	96.6	98.0	98.8
Conformal LR	98.6	99.8	98.4	96.4	97.8	97.6
Conformal NRE-A	98.6	99.8	98.4	96.3	97.9	97.6
Conformal NRE-C	98.6	99.9	98.4	96.4	98.2	97.7
$K = 1$ null simulation						
Conformal LR	99.3	100.0	99.1	97.5	99.1	98.5
Conformal NRE-A	99.3	100.0	99.0	97.7	99.2	98.3
Conformal NRE-C	99.3	100.0	98.9	97.6	99.1	98.2
Model-based plug-in alternative						
Direct LR	98.3	99.9	98.6	96.7	98.5	99.2
Conformal LR	98.7	99.9	98.0	97.0	98.0	97.5
Conformal NRE-A	98.7	99.9	98.0	97.1	97.9	97.8
Conformal NRE-C	98.5	99.8	97.9	97.1	98.0	97.8
Small architecture						
Conformal NRE-A	99.7	100.0	99.8	100.0	100.0	100.0
Conformal NRE-C	98.8	100.0	98.8	96.5	97.9	97.9

of width 8. In every case coverage remains valid; a misspecified score forfeits power, never validity, and NRE-C's contrastive objective is markedly more robust to this misspecification.

F.2.3 Per-ablation confidence sequences ($\theta_* = 0$)

Table 2 records each procedure at two time points; here we show the full sequential trajectories at $\theta_* = 0$, for both models, with the infeasible oracle variant of every estimator (which uses θ_* as the alternative) shown dashed. Figure 13 shows the baseline: the four feasible curves (exact-LR and neural) are essentially superimposed in every panel, with miscoverage plateauing well below α, coincident hull widths, and final-time retention concentrating similarly at θ_*, so the choice of score is invisible on these models. The oracle variants never miscover and give narrower sets at every time, so the extra width of the feasible sets comes from estimating the alternative rather than from the score.

Replacing minimum evidence with the model-based running estimate (Fig. 14) separates the two models: the Gaussian trajectories are indistinguishable from the baseline, while on g-and-k the feasible width curves sit visibly above both the baseline and the oracle, with final-time widths roughly twice the baseline (Table 2); retention concentrates correspondingly less sharply. Miscoverage is unaffected in both cases, as validity does not depend on the alternative.

Reducing the null budget to $K = 1$ (Fig. 15) changes only the conformal sequences, which use the K reference draws; the direct sequences are unchanged. The conformal widths inflate (note the rescaled axis), the three scores degrading together, and the miscoverage of the feasible conformal sequences rises toward the nominal line. Validity is retained throughout.

Underparameterizing the score network (Fig. 16) degrades NRE-A on both models while leaving NRE-C close to its baseline. The degradation is mild on the Gaussian (NRE-A's sets widen but still concentrate) and severe on g-and-k, where its width stays close to the full grid and its retention never concentrates. Miscoverage nonetheless stays below α for every curve: the underparameterized score forfeits power, sharply so on g-and-k, without ever compromising coverage.

F.3 SLCP

Figure 17 adds conformal NRE-A and the oracle versions of the conformal exact-LR, NRE-A and NRE-C sequences to Fig. 2; the set-up is described in Section E.

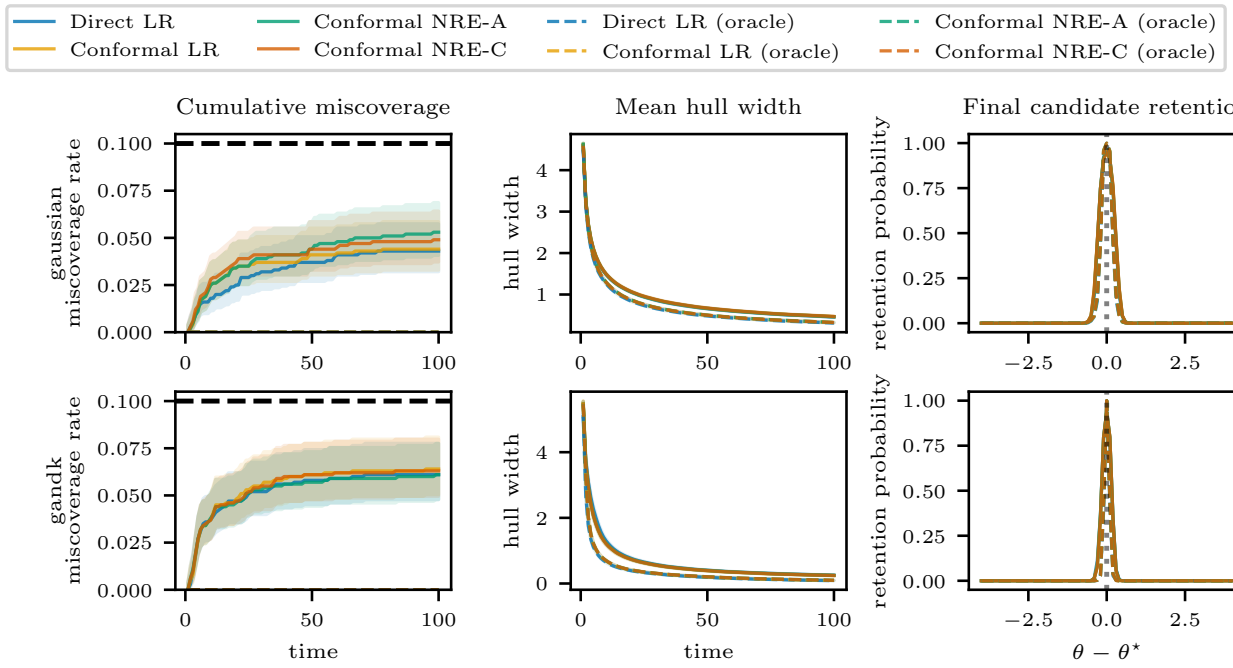


Figure 13: Sequential confidence sequences at $\theta_* = 0$ for the Gaussian (top) and g-and-k (bottom) models under the baseline configuration (minimum-evidence alternative, $K = 15$). Left: cumulative miscoverage (dashed $\alpha = 0.1$). Center: mean hull width of nonempty sets. Right: final-time retention per candidate. Solid: feasible procedures; dashed: infeasible oracle using θ_* as the alternative.

Cumulative miscoverage stays below α for every procedure. The three conformal sequences shrink at similar rates, NRE-A slightly more slowly, and direct LR shrinks fastest. At $n = 100$ they retain between 4.1 and 8.2 candidates on average, against about 4 for the oracles: θ_* and its three mirror images, which give the same distribution of the data. Table 4 shows which candidates remain. The exact-LR methods retain or exclude the four equivalent values together, since their likelihood ratios and shared reference draws yield identical evidence processes; with θ_* as the alternative, they therefore never exclude any of the four. Neural scores need not preserve this symmetry: conformal NRE-C retains all four in 93.4% of repetitions, close to its coverage of θ_* (94.6%), but its final set consists of exactly these four in only 28.2%, with 2.56 other candidates retained on average, and the neural oracle variants exclude at least one of the three other values in 13–17% of repetitions while always retaining θ_*. The coverage guarantee applies to each parameter value taken as the truth, not to all four jointly.

G APPLICATION: AUDITING MODEL ROUTING IN AN LLM API

G.1 Setting and data

The audit problem. An API that advertises a language model P may serve a different one without saying so [16, 4], for example by answering a fraction θ of the queries with a cheaper model R [4, 55]. An auditor who can only send queries and read the replies wants a confidence sequence for θ.

Our setup follows IRIS [55], a recent black-box auditor for this problem. It sends prompts that ask for random output, marks the replies that look atypical for the advertised model, and detects routing and estimates θ from how often such replies occur. The approach hinges on the fact that language models are poor random number generators, and each model is poor in its own reproducible way. We use IRIS's prompts, its parser and its request settings, and in Section G.6 we compare with its fixed-time confidence interval for θ and with a confidence-sequence version of its sequential test.

Probes. We use two of IRIS's prompts as probes to detect routing anomalies. The prompt `short_digits_8` asks "Generate exactly 8 independent uniformly random digits from 0 to 9. Output only the compact digit

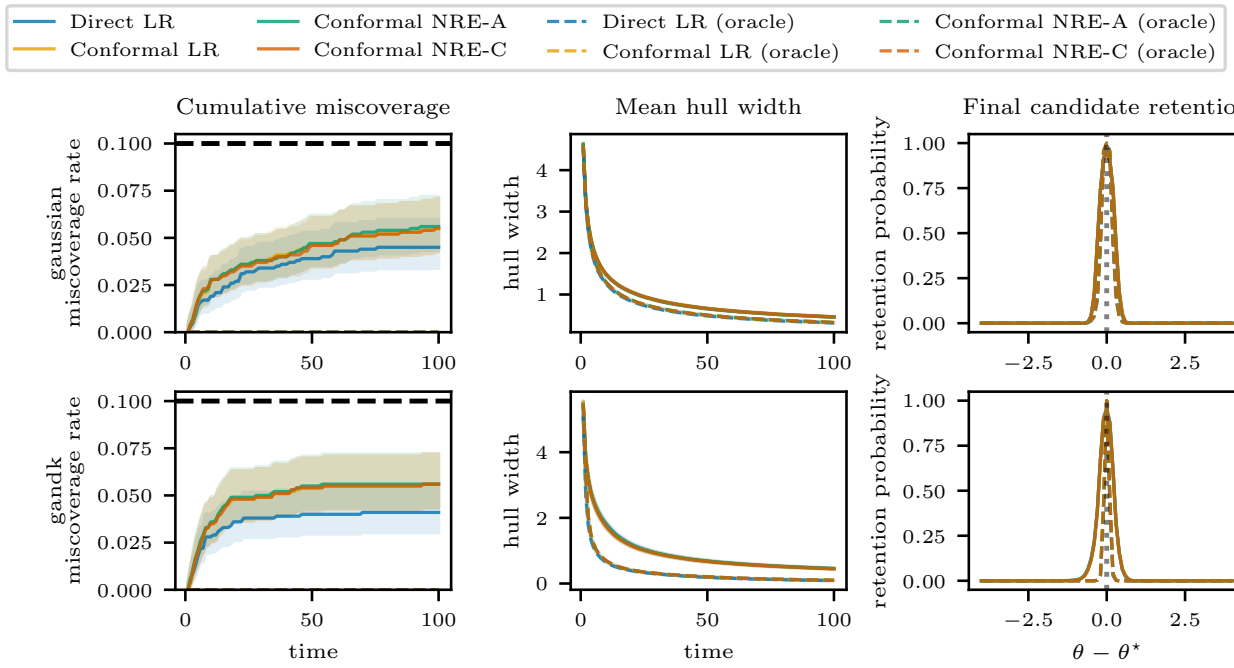


Figure 14: As Fig. 13, with the model-based plug-in (running mean / running median) in place of minimum evidence.

string, no separators.", and `single_number_100` asks "Choose one uniformly random integer between 1 and 100 inclusive. Output only the integer." IRIS's parser reduces each reply to a symbol y: the string of digits for the first prompt, and the first integer in the reply for the second.

Data. We take P = `qwen3:8b` and R = `qwen3:1.7b` [54], served locally by Ollama[1] with IRIS's request settings: reasoning off, temperature 1, top-p 1, and IRIS's limit on reply length. We collected 28,000 replies per model and prompt. Each query is a separate call with the same prompt and no conversation history, so the replies of one model are i.i.d. draws from its output distribution given the prompt.

On both prompts the two models' replies barely overlap: each model's most frequent reply is one that the other model almost never gives (Table 5), so a single reply is strong evidence of which model answered. On the number prompt the two models are separated by the wording of the reply rather than by the number chosen: `qwen3:1.7b` restates the task before answering, so the parser records 1. The prompts also differ in how many distinct replies occur in our collections: 1,008 for the digit prompt and 20 for the number prompt.

Semi-synthetic design. As in IRIS's routing experiments, we build every audited stream ourselves by simulating the routing while using real LLM responses. Given $\theta_* \in [0, 1]$, each query is answered with a reply from R or P by sampling without replacement from a held-out set of the models' collected replies with probabilities θ_* and $(1 - \theta_*)$, respectively. The replies are real outputs of the two models; only the routing is simulated. Because θ_* is known, this lets us measure the coverage of every method.

[1] `https://ollama.com`, version 0.34.4.
[2] Ten digits, although the prompt asks for eight.
[3] `qwen3:1.7b` opens its reply with "Let's choose a random integer between 1 and 100 ...", and the parser keeps the first integer in the reply.

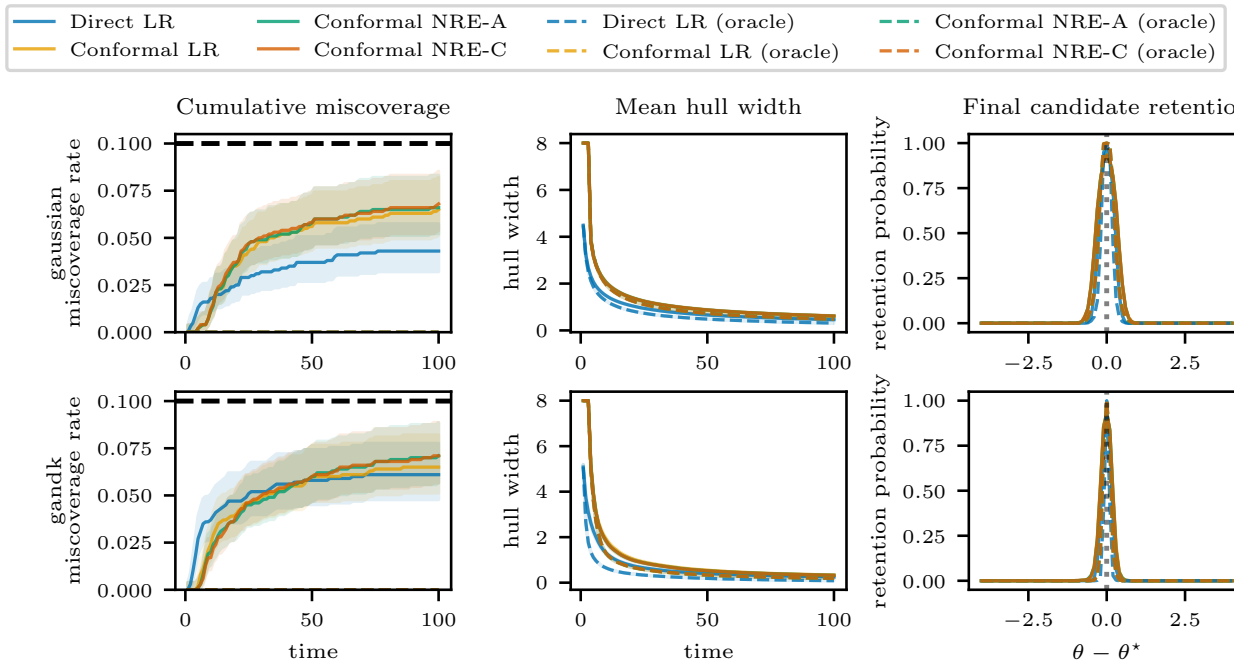


Figure 15: As Fig. 13, with $K = 1$ null simulation per step.

Table 4: SLCP candidate retention at $n = 100$ over 1,000 repetitions, discarding the first two e-values (Section E.3). All four: percentage of repetitions retaining all four equivalent values $(0, 0, \pm 2.2, \pm 2.2, 0)$. Other candidates: mean number of retained candidates outside these four, over all repetitions. Exactly four: percentage of repetitions whose final set consists of precisely these four values.

Method	All four (%)	Other candidates	Exactly four (%)
Direct LR	96.1	0.23	93.3
Conformal LR	93.2	1.78	53.5
Conformal NRE-A	92.9	4.38	20.8
Conformal NRE-C	93.4	2.56	28.2
Direct LR (oracle)	100.0	0.00	100.0
Conformal LR (oracle)	100.0	0.00	100.0
Conformal NRE-A (oracle)	83.5	0.28	67.0
Conformal NRE-C (oracle)	87.1	0.03	84.4

G.2 Simulator

Let p and r be the pmf of the parsed reply y under P and R, respectively. At routing fraction θ, the simulator sends a query to R with probability θ and to P otherwise, so its replies are i.i.d. with pmf

$$f_\theta(y) = (1-\theta)\, p(y) + \theta\, r(y), \qquad \theta \in [0,1]. \tag{50}$$

In practical applications, the likelihood f_θ has no closed form: p and r are the models' output pmfs over parsed replies, which are not exposed by APIs in the audit setting. The auditor can only sample from them by querying the models.

For any $\theta_0, \theta_1 \in [0,1]$, the likelihood ratio depends on the reply only through $\rho(y) = r(y)/p(y)$:

$$\frac{f_{\theta_1}(y)}{f_{\theta_0}(y)} = \frac{(1-\theta_1) + \theta_1 \rho(y)}{(1-\theta_0) + \theta_0 \rho(y)}. \tag{51}$$

Hence an estimate of ρ, learned from replies of the two models, gives an estimated likelihood ratio for every pair of candidates θ_0, θ_1. Our audited streams are built from held-out replies of the two models, with routing that we apply ourselves.

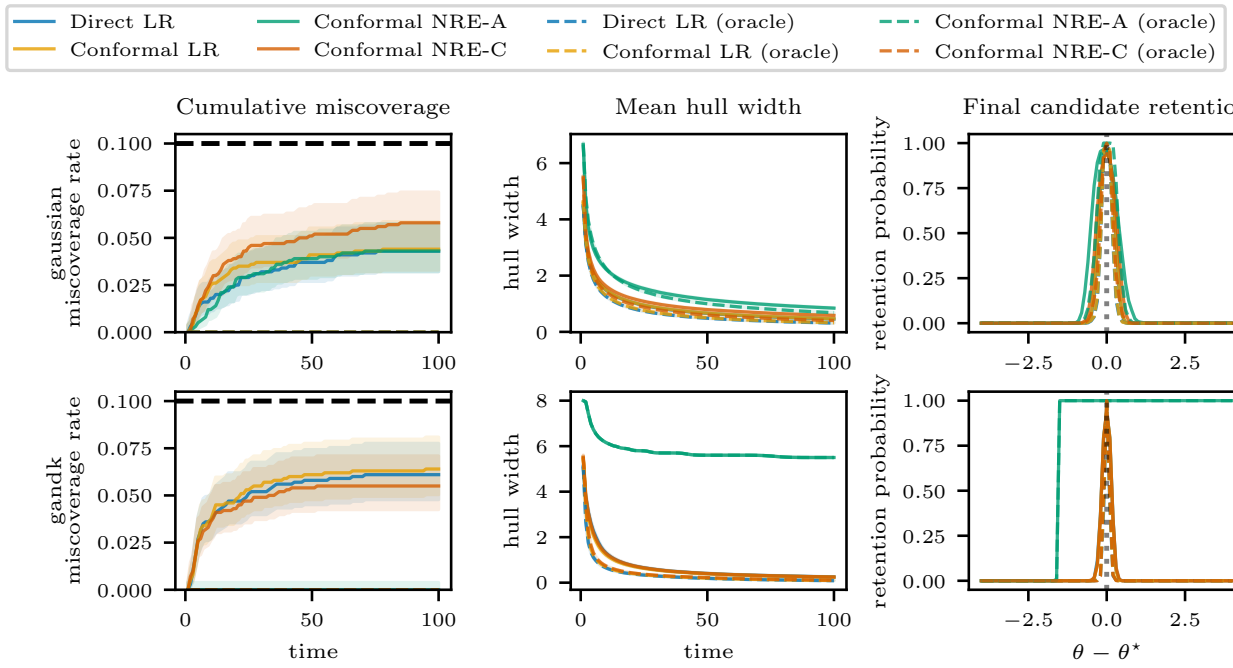


Figure 16: As Fig. 13, with a deliberately underparameterized score network (single hidden layer of two units).

Table 5: The most frequent parsed reply of each model, and how often each model gives it, out of 28,000 calls per model and prompt.

Prompt	Parsed reply	`qwen3:8b` (P)	`qwen3:1.7b` (R)
8 random digits	`73951842`	29%	never
8 random digits	`0123456789`[2]	never	95%
a number from 1 to 100	42	92%	9 replies
a number from 1 to 100	1[3]	never	99.9%

G.3 Computing the scores

Of the 28,000 replies collected per model and prompt, we set aside 10,000 of P's and 10,000 of R's to estimate the scores and use them for nothing else; we call these the score pools. The remaining replies are used to build the audited streams and their references (Section G.4). From the score pools we estimate the ratio $\rho(y) = r(y)/p(y)$ in one of two ways.

Count ratio. Since the number of unique responses given by the LLMs are quite low for this experiment, it is feasible to estimate $p(y)$ and $r(y)$ by the share of each score pool's replies equal to y, with half a count added to every possible reply. The latter smoothing keeps $\hat{\rho}$ finite and positive, and a reply seen in neither pool gets $\hat{\rho} = 1$, so it carries no evidence.

NRE. A binary NRE is trained to tell P's replies from R's. The network has two hidden layers of 64 units. Its inputs are features of the reply string: its length, the frequency of each character, the first and last characters, how often consecutive characters repeat or step by one, and indicators for the 64 most frequent replies. It is trained with a class-balanced logistic loss, so its output estimates $\log \rho(y)$. An amortised NRE, as in the other experiments, is not needed here, because θ enters only through Eq. (51).

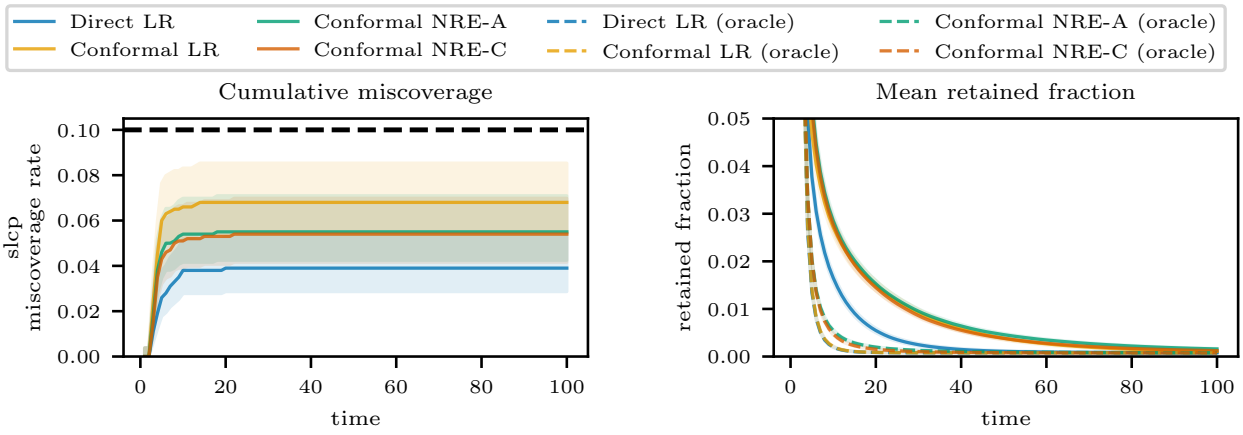


Figure 17: Sequential confidence sequences for the SLCP model, all procedures. Left: cumulative miscoverage (Wilson 95% bands; dashed $\alpha = 0.1$; oracles omitted). Right: mean fraction of candidates retained. Solid: feasible procedures; dashed: infeasible oracle using θ_* as the alternative.

Score. With either estimate $\hat{\rho}$, plugging it into Eq. (51) gives an estimate of the likelihood ratio between any two candidates. The score of candidate θ at step i is

$$T_i^{(\theta)}(y) = \frac{\hat{f}_{\hat{\theta}}(y)}{\hat{f}_{\theta}(y)}, \tag{52}$$

where $\hat{\theta} = \hat{\theta}_{i-1}$ is the minimum-evidence alternative: the candidate with the least accumulated evidence against it after the first $i-1$ observations.

Defensive score. We use $(1-\lambda)\,T_i^{(\theta)}(y) + \lambda$ with $\lambda = 0.3$ in place of $T_i^{(\theta)}(y)$. Mixing every score with the same constant brings the scores of different replies closer together, so each step changes a candidate's wealth less. This matters early in the sequence. The alternative $\hat{\theta}$ starts at 0 and moves towards the true θ_* only as evidence accumulates, and while it is still far away, every candidate between it and the truth loses wealth, and those that should eventually be excluded must later win it back. The mixed score is still positive and depends only on past observations, so validity holds for any fixed λ; only efficiency depends on it. We chose λ on held-out development streams, by a rule fixed before running them: with the count-ratio score, both prompts, $\lambda \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5\}$ and $\theta_* \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5\}$, take the value with the smallest mean hull width at $n = 300$ among those whose cumulative miscoverage is at most α up to its Wilson band and whose median detection time is at most one query slower than with $\lambda = 0$, at every θ_*. The chosen $\lambda = 0.3$ is then used unchanged for every score and prompt, and every result we report comes from fresh test streams, drawn independently of the development streams. On the test streams (Table 6), $\lambda = 0.2$ and 0.3 narrow the final sets by about 30% relative to $\lambda = 0$ without slowing detection; $\lambda = 0.5$ can slow detection when routing is rare. Discarding the first few observations instead narrows the sets only at large θ_* and delays detection, and the average of the $\lambda = 0, 0.2, 0.5$ e-processes, which involves no tuning, comes within a few percent of the chosen value.

G.4 Experiments

The 28,000 replies of each model per prompt are split into disjoint pools: 10,000 form the score pool (Section G.3), 3,000 an observation pool from which the streams are built, and the remaining 15,000 five reference pools of 3,000 from which the references are drawn.

At step i, the confidence sequence takes K fresh replies $R_{i1}, \ldots, R_{iK}$ of R and $P_{i1}, \ldots, P_{iK}$ of P from a reference pool, together with uniform draws $U_{i1}, \ldots, U_{iK}$, and sets, for every candidate θ,

$$\widetilde{Y}_{ij}^{(\theta)} = \begin{cases} R_{ij} & U_{ij} < \theta, \\ P_{ij} & \text{otherwise.} \end{cases} \tag{53}$$

Table 6: Sensitivity of the routing confidence sequence (count-ratio score) to the defensive floor λ, on fresh test streams disjoint from those used to choose it (200 repetitions per $\theta_* \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5\}$, $n = 300$, $\alpha = 0.1$). Miscov. is the largest time-uniform miscoverage over θ_*; width is the mean hull width at $n = 300$ over nonempty sets; detect is the median number of queries until $\theta = 0$ leaves the set. "Skip k" discards the first k observations, so its guarantee holds from step $k + 1$; the mixture averages the $\lambda = 0, 0.2, 0.5$ e-processes and involves no tuning. Every miscoverage is at most α up to its Wilson 95% band; $\lambda = 0.3$ is the value used elsewhere, chosen on separate development streams.

		width at $\theta_* =$					detect at $\theta_* =$				
Variant	miscov.	0.05	0.1	0.2	0.3	0.5	0.05	0.1	0.2	0.3	0.5
8 random digits											
$\lambda = 0$	0.105	0.087	0.129	0.197	0.230	0.278	32	15	8	6	3
$\lambda = 0$, skip 2	0.105	0.083	0.116	0.169	0.179	0.188	33	15	8	6	4
$\lambda = 0$, skip 5	0.085	0.076	0.104	0.142	0.149	0.163	32	17	10	8	7
$\lambda = 0.2$	0.080	0.061	0.091	0.138	0.163	0.194	30	15	8	5	3
$\lambda = 0.3$	0.080	0.059	0.089	0.133	0.156	0.186	30	15	8	5	3
$\lambda = 0.5$	0.055	0.062	0.092	0.135	0.160	0.184	42	18	10	6	3
mixture of $\lambda \in \{0, 0.2, 0.5\}$	0.080	0.064	0.093	0.140	0.164	0.193	32	15	8	6	3
a number from 1 to 100											
$\lambda = 0$	0.100	0.087	0.134	0.205	0.243	0.289	30	15	8	6	3
$\lambda = 0$, skip 2	0.090	0.083	0.123	0.174	0.185	0.193	30	15	8	6	4
$\lambda = 0$, skip 5	0.085	0.077	0.107	0.142	0.150	0.163	30	16	10	8	7
$\lambda = 0.2$	0.095	0.059	0.091	0.140	0.163	0.197	30	15	8	6	3
$\lambda = 0.3$	0.080	0.058	0.089	0.134	0.160	0.187	30	15	8	6	3
$\lambda = 0.5$	0.050	0.060	0.090	0.134	0.161	0.184	30	15	8	6	3
mixture of $\lambda \in \{0, 0.2, 0.5\}$	0.080	0.061	0.094	0.141	0.165	0.195	30	15	8	6	3

Since θ only decides which model answers, these are draws from f_θ for every candidate at once. All candidates share the same $2K$ replies and K uniform draws (common random numbers), so the number of model calls does not grow with the number of candidates.

A stream of $n = 300$ steps uses $K = 10$ replies of each model per step, so exactly one reference pool, and no reply is used twice within a repetition. As is usual in semi-synthetic designs, replies are reused across repetitions: each reference pool serves 40 of the 200 repetitions, reshuffled each time, and all streams draw from the same observation pool.

For each prompt and each true routing fraction $\theta_* \in \{0, 0.1, 0.3\}$, we run 200 repetitions. We compare three confidence sequences:

- *Conformal count ratio.* Count-ratio score, minimum-evidence alternative.
- *Conformal NRE.* NRE score, minimum-evidence alternative.
- *Conformal count ratio (oracle).* Count-ratio score, with the true θ_* as the alternative.

For each method we report the following quantities:

- *Cumulative miscoverage.* The fraction of repetitions in which the confidence sequence has excluded the true θ_* at some step $i \leq n$. Figures show it with Wilson 95% intervals.
- *Mean hull width.* The largest retained candidate minus the smallest, averaged over the repetitions with a nonempty set.
- *Detection time.* The number of steps until $\theta = 0$ leaves the set; we report its median, because it is infinite in repetitions where $\theta = 0$ is never excluded.
- *Confidence sets over time.* For every θ_* we also show the sets themselves. At each step n we take the lower and upper ends of each repetition's accepted candidate set, over the repetitions with a nonempty set, and plot the mean of the lower ends and the mean of the upper ends, with 95% confidence intervals for the means.

G.5 Baselines

All baselines are computed on the same streams as the conformal confidence sequences, over the same 101 candidates.

Fixed-n baseline: IRIS interval. IRIS [55] reduces each reply to a single bit, a *tell*, which marks the reply as atypical for the advertised model P. In IRIS, a reply y is a tell when $-\log \hat{\mathbb{P}}(P \mid y)$, computed from a classifier that predicts which model produced y, exceeds a high quantile of the same score on P's replies. We apply the same rule to our count-ratio estimate: a reply is a tell if $\log \hat{\rho}(y)$ exceeds a threshold chosen so that at most 5% of P's replies in the score pool are tells. Let a_1 and q be the shares of tells among P's and R's replies in the score pools (with half a count added). As in IRIS, the threshold, a_1 and q are fixed before the audit starts; during the audit, each reply is only checked against the threshold. Under routing fraction θ, a reply is a tell with probability $(1-\theta)a_1 + \theta q$. If f_n is the share of tells among the first n replies, IRIS's estimate is $\hat{\theta}_n = (f_n - a_1)/(q - a_1)$, clipped to $[0,1]$, and its delta-method interval accounts for the randomness in f_n and for the estimation of a_1 and q on the score pools. The interval is built for a single n fixed in advance, and even there its coverage is nominal only asymptotically, because the delta method relies on a normal approximation. With very few replies the interval is degenerate. We therefore count it as covering θ_* for $n < 10$, which favours it, and in the figures we plot its lower and upper ends only from $n = 2$.

Anytime-valid baseline: Betting IRIS. IRIS [55] gives a confidence interval for θ only at a fixed n, and uses a betting test on the tells only to decide whether the endpoint is honest. We extend this test to a confidence sequence for θ [52]. Let X_i be the tell indicator of reply i, defined as in the IRIS interval above. Under routing fraction θ, the tell rate is $p_\theta = (1-\theta)a_1 + \theta q$, so

$$Z_i^{(\theta)} = X_i - p_\theta \in [-1, 1] \tag{54}$$

has mean zero when θ is the true routing fraction. For each candidate θ we track the wealth

$$W_n^{(\theta)} = \tfrac{1}{2}\prod_{i=1}^{n}\bigl(1 + \lambda_i^{+} Z_i^{(\theta)}\bigr) + \tfrac{1}{2}\prod_{i=1}^{n}\bigl(1 - \lambda_i^{-} Z_i^{(\theta)}\bigr) \tag{55}$$

of two bettors, one on a positive and one on a negative mean of $Z_i^{(\theta)}$, who start with half the wealth each. The bets $\lambda_i^{\pm} \in [0, 0.9]$ are chosen from $Z_1^{(\theta)}, \ldots, Z_{i-1}^{(\theta)}$ by the aGRAPA rule of Waudby-Smith and Ramdas [52], and θ is excluded from the set once $W_n^{(\theta)}$ reaches $1/\alpha$. The sequence is only approximately valid, because a_1 and q are estimated on the score pools, so p_θ is not exactly the tell rate under θ.

G.6 Results

Validity. The conformal sequences and Betting IRIS keep cumulative miscoverage below $\alpha = 0.1$ throughout, at every θ_* (Table 7; Figs. 18 to 20, panels a and c). The IRIS interval, checked at every step, does not: its cumulative miscoverage exceeds α at $\theta_* = 0.1$ and 0.3 on both prompts, and at $\theta_* = 0$ on the digits prompt.

Width. The conformal sequences are narrower than Betting IRIS throughout at $\theta_* = 0$ and 0.1, and early on at $\theta_* = 0.3$, where Betting IRIS ends slightly narrower, with the NRE matching the count ratio (Table 7; Figs. 18 to 20, panels b and d). The sets are nonempty in almost every repetition (Table 8). The conformal sequence with the oracle alternative θ_* achieves the smallest width among the confidence sequences, suggesting that further gains may be obtained by tuning $\hat{\theta}$.

Detection. The conformal sequences with the count ratio and the NRE rule out honest service, $\theta = 0$, four to five times sooner than Betting IRIS, at both $\theta_* = 0.1$ and 0.3 (Table 7). With the oracle alternative, the count-ratio sequence is more than twice as fast again, so here too a better choice of $\hat{\theta}$ could help.

G.7 Limitations

The simulator assumes that every reply comes from P or R, run with the settings used to collect our pools, and that each query is routed independently. If the API departs from this, for example by also using a third model,

Table 7: LLM routing audit on semi-synthetic test streams, drawn independently of the development streams used to choose $\lambda = 0.3$: 200 repetitions for each $\theta_* \in \{0, 0.1, 0.3\}$, $n = 300$ steps, $\alpha = 0.1$. Miscoverage is the cumulative miscoverage at $n = 300$, taking the largest of the three values of θ_*; for the IRIS interval it counts misses from $n = 10$ onwards. Width is the mean hull width at $n = 300$ over repetitions with a nonempty set, at $\theta_* = 0$, 0.1 and 0.3. Detect is the median number of steps until $\theta = 0$ leaves the set, at $\theta_* = 0.1$ and 0.3.

	8 random digits						a number from 1 to 100					
		width			detect			width			detect	
Method	miscov.	0	0.1	0.3	0.1	0.3	miscov.	0	0.1	0.3	0.1	0.3
Conformal confidence sequences												
Conformal count ratio	0.08	0.01	0.09	0.16	16	5	0.09	0.01	0.09	0.16	16	5
Conformal NRE	0.08	0.01	0.09	0.16	17	5	0.09	0.01	0.09	0.16	17	5
Conformal count ratio (oracle)	0.00	0.01	0.07	0.11	7	2	0.00	0.01	0.07	0.11	7	2
Anytime-valid baseline												
Betting IRIS	0.06	0.05	0.11	0.14	73	21	0.07	0.03	0.10	0.13	81	23
Fixed-n baseline												
IRIS interval	0.70	0.03	0.06	0.08	21	4	0.66	0.02	0.06	0.08	24	4

Table 8: Percentage of nonempty confidence sets at $n = 300$, over 200 repetitions. Mean hull widths in Table 7 are averaged over these repetitions only.

	8 random digits			a number from 1 to 100		
Method	$\theta_* = 0$	$\theta_* = 0.1$	$\theta_* = 0.3$	$\theta_* = 0$	$\theta_* = 0.1$	$\theta_* = 0.3$
Conformal confidence sequences						
Conformal count ratio	100.0	100.0	100.0	100.0	100.0	100.0
Conformal NRE	100.0	100.0	100.0	100.0	100.0	100.0
Conformal count ratio (oracle)	100.0	100.0	100.0	100.0	100.0	100.0
Anytime-valid baseline						
Betting IRIS	99.0	100.0	100.0	94.0	100.0	100.0
Fixed-n baseline						
IRIS interval	100.0	100.0	100.0	100.0	100.0	100.0

our scores have little power against it: they are likelihood ratios between members of the model $\{f_\theta\}$, so they only compare values of θ. Detecting such departures would need a score that targets them.

H APPLICATION: THE LENS MASS FUNCTION FROM MICROLENSING TIMESCALES

H.1 Setting and data

Scientific target. We infer the mass function of stars and brown dwarfs toward the centre of the Milky Way, that is, how common objects of each mass are. It is modelled as a power law $\xi(M) \propto M^\alpha$ whose slope changes at a few fixed masses. The parameter of interest is $\theta = (\theta_1, \theta_2)$: the slope θ_1 for low-mass stars and the slope θ_2 for brown dwarfs. These objects are faint, and brown dwarfs emit almost no light, so they are hard to count directly. Microlensing detects them through their gravity instead, whether or not they shine.

Microlensing events. When such an object (the *lens*) passes almost exactly in front of a more distant star (the *source*), its gravity bends the source's light, and the source brightens and then fades back over a few days to a few months. This brightening is a microlensing event [32]. OGLE (the Optical Gravitational Lensing Experiment), a long-running survey with a dedicated telescope in Chile, detects such events by repeatedly photographing the same patches of sky toward the centre of the Milky Way and recording the brightness of millions of stars.

The length of an event, its Einstein timescale t_E, is longer for heavier lenses. Because t_E also depends on the lens's and source's distances and speeds, one event says little about the lens's mass. The distribution of t_E over many events, however, depends on how common lenses of each mass are, and hence on θ. Section H.2 gives this

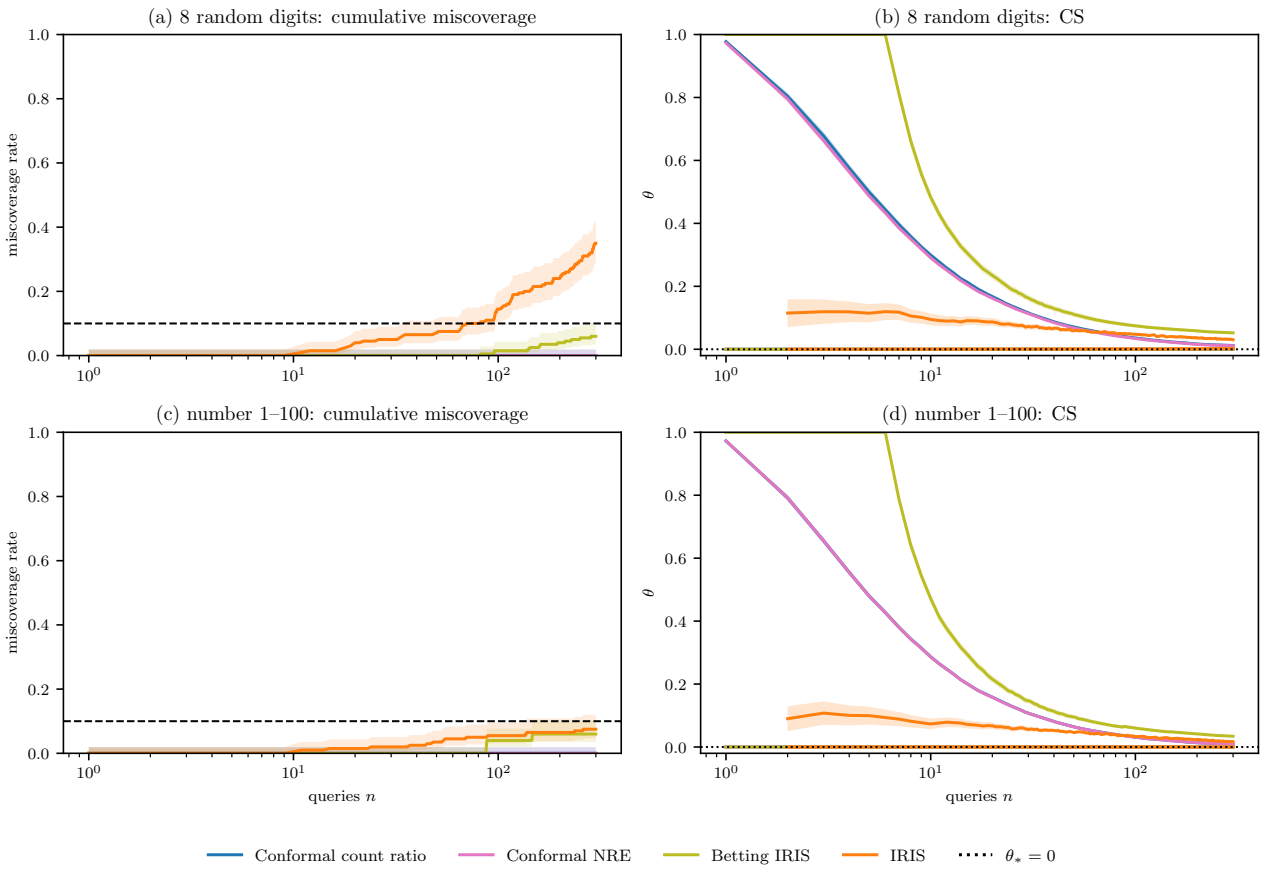


Figure 18: LLM routing audit at $\theta_* = 0$ (an honest API), one row per prompt. (a, c) Cumulative miscoverage (Wilson 95% bands; dashed $\alpha = 0.1$); the fixed-time IRIS interval is counted as covering θ_* for $n < 10$. (b, d) Confidence sequences: mean lower and upper ends over the 200 repetitions with a nonempty set, with 95% intervals for the means; IRIS is shown from $n = 2$.

distribution for each θ, given the event's field and measurement error.

Data. We use the catalogue of Mróz et al. [31]: 5,790 events detected in the low-cadence fields of the fourth phase of OGLE (OGLE-IV) over eight yearly seasons (2010–2017). The survey divides the sky into fixed patches called *fields*. For each event the catalogue gives its field, its estimated timescale $\hat{t}_{E,i}$, and error bars e_i^+ and e_i^- on that estimate. For each field f it also gives the detection efficiency $\varepsilon_f(t_E)$, the probability that an event of timescale t_E in that field is detected; short events are easily missed between photographs, so ε_f is small at short t_E. We keep the 3,514 events in the 70 fields with Galactic coordinates $|\ell| \le 10^\circ$ and $3^\circ \le |b| \le 10^\circ$, roughly the region covered by the simulator.

Observations and covariates. Each event i gives an observation $Y_i = \log_{10} \hat{t}_{E,i}$ and covariates $X_i = (\ell_{f_i}, b_{f_i}, \sigma_i)$: the coordinates of the centre of its field and its measurement error on the log scale, $\sigma_i = \frac{1}{2}[\log(1 + e_i^+/\bar{t}_{E,i}) - \log(1 - e_i^-/\bar{t}_{E,i})]$, where $e_i^\pm \ge 0$ are the catalogue's error bars about its median timescale $\bar{t}_{E,i}$. We model Y_i given X_i: for a given field, the simulator (Section H.2) produces the event's true timescale t, and the measured timescale is $\hat{t}_{E,i} = t\, e^{\sigma_i \eta}$ with $\eta \sim \mathcal{N}(0,1)$. This symmetric log-normal error, with σ_i treated as known, approximates the catalogue's asymmetric error bars. The references for event i are therefore simulated in the same field and with the same error (Section H.2).

Independence. Microlensing events occur independently of one another, at random times and places, and each involves a different pair of lens and source. The survey detects each event with a probability that depends only on its field and timescale. Given their number, the detected events are therefore independent draws from

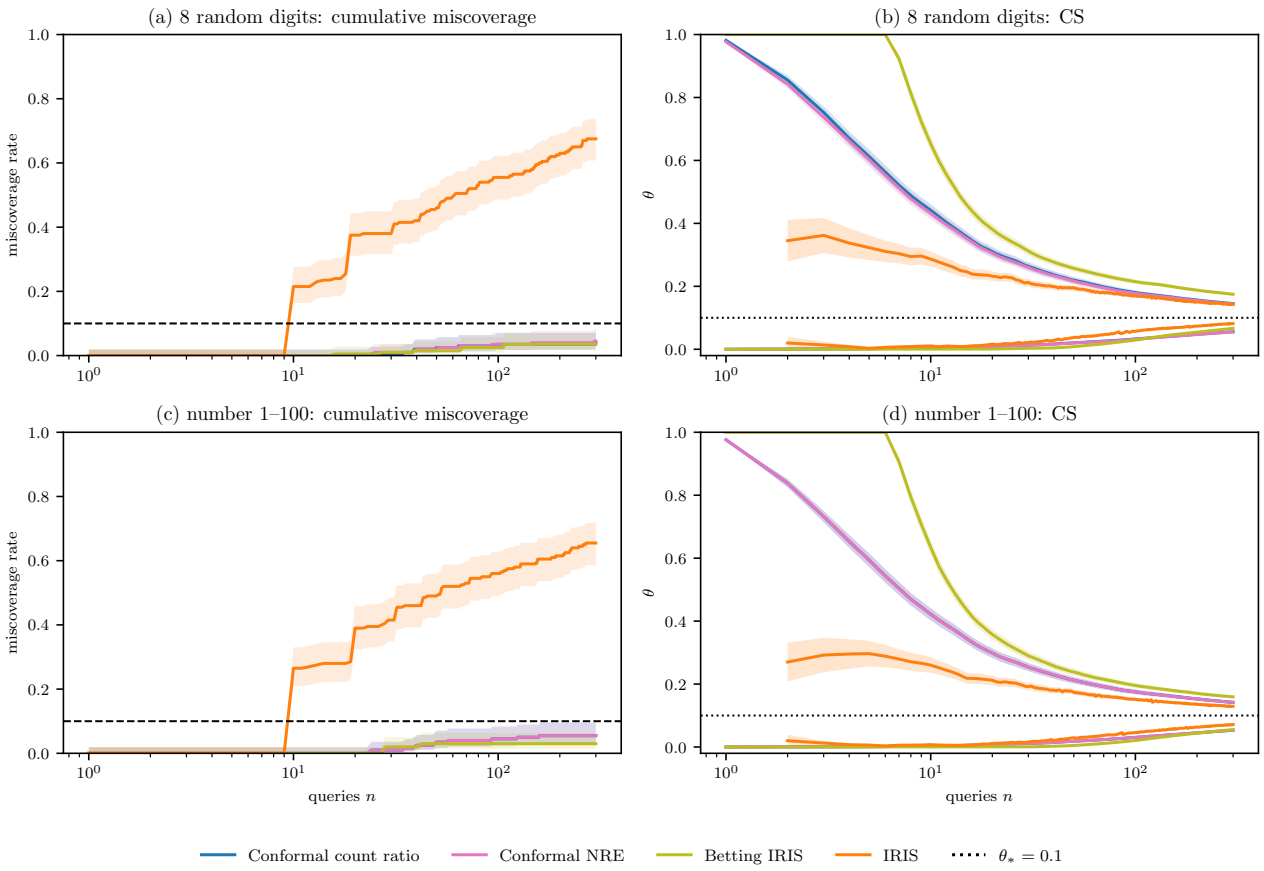


Figure 19: LLM routing audit at $\theta_* = 0.1$, one row per prompt. (a, c) Cumulative miscoverage (Wilson 95% bands; dashed $\alpha = 0.1$); the fixed-time IRIS interval is counted as covering θ_* for $n < 10$. (b, d) Confidence sequences: mean lower and upper ends over the 200 repetitions with a nonempty set, with 95% intervals for the means; IRIS is shown from $n = 2$.

one distribution, so the pairs (X_i, Y_i) are i.i.d.; in particular, given the covariates, the timescales are independent and each Y_i depends only on its own X_i, which is what the conditional construction requires (Theorem C.1). Each repetition of the confidence sequence processes the events in a fresh random order.

Held-out and analysed events. We split the events by season. The 1,716 events from even seasons (2010, 2012, 2014 and 2016) are held out and used only to check the model (Section H.4) and to choose the truth of the semi-synthetic runs. The 1,798 events from odd seasons (2011, 2013, 2015 and 2017) are the data on which the confidence sequence is run, and on which the fitted model is tested.

H.2 Simulator

Galactic model. We use the Galactic model of Koshimoto et al. [26], fitted to Gaia, VVV, BRAVA, and OGLE data, as implemented in `genulens` [25]. It describes the Galaxy as different groups of stars, or *components*, each with its own spatial density, velocities, and age; lenses and sources are drawn from these components.

Parameter. The parameter $\theta = (\theta_1, \theta_2)$ enters the simulator through the initial mass function $\xi_\theta(M)$, the number of objects per unit mass with mass M at birth. It is a broken power law. The mass range from 0.001 to $120\,M_\odot$ is divided at 0.01, 0.08, and $0.86\,M_\odot$ into four intervals, and on the k-th interval $\xi_\theta(M) \propto M^{\alpha_k}$, where $\alpha_2 = \theta_1$ and $\alpha_3 = \theta_2$ are the free slopes and α_1 and α_4 are fixed; on a log–log plot, ξ_θ is a straight line within each interval, and the pieces join at the break masses. From heaviest to lightest, the intervals and their slopes are:

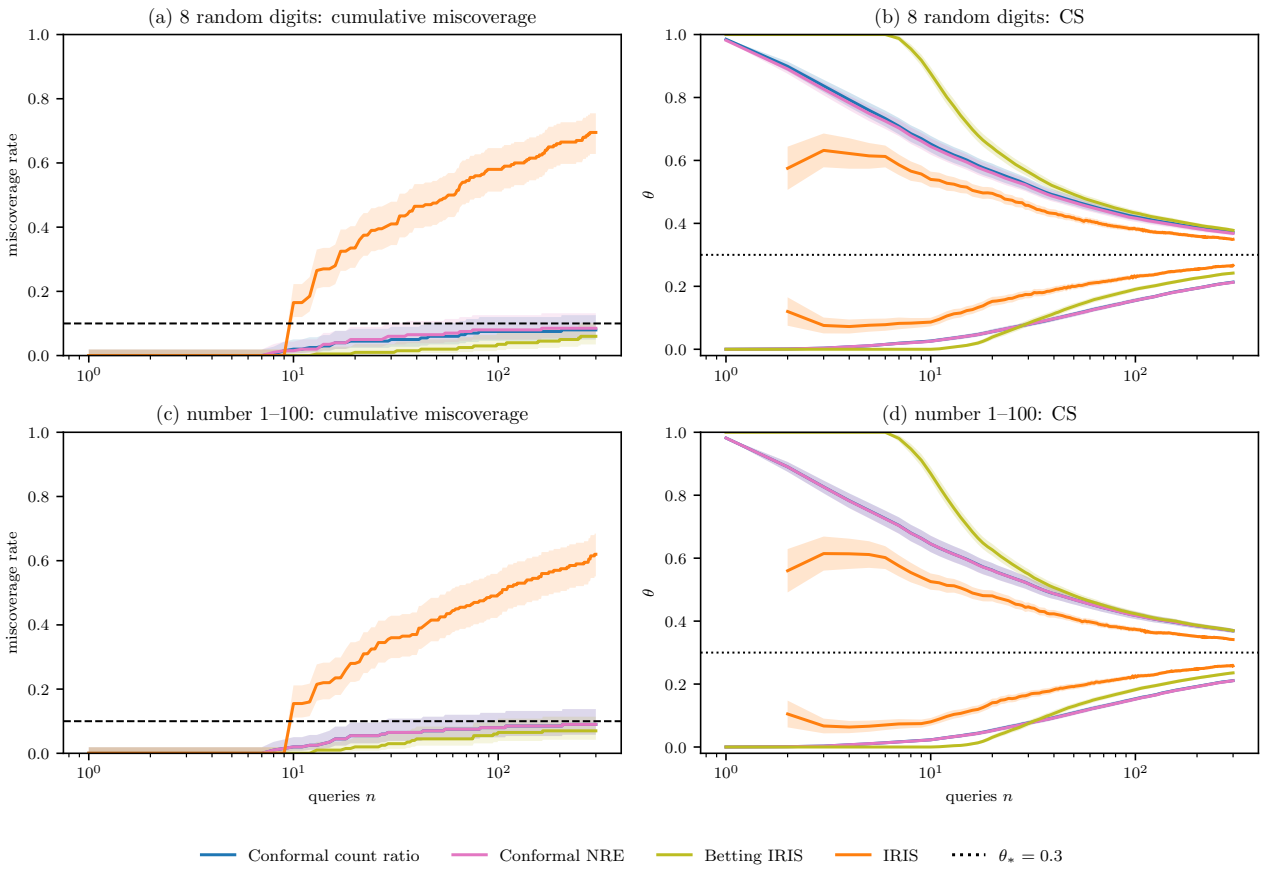


Figure 20: LLM routing audit at $\theta_* = 0.3$, one row per prompt. (a, c) Cumulative miscoverage (Wilson 95% bands; dashed $\alpha = 0.1$); the fixed-time IRIS interval is counted as covering θ_* for $n < 10$. (b, d) Confidence sequences: mean lower and upper ends over the 200 repetitions with a nonempty set, with 95% intervals for the means; IRIS is shown from $n = 2$.

Mass range ($M_\odot$)	Objects	Slope
0.86–120	heavier stars	$\alpha_1 = -2.32$ (fixed)
0.08–0.86	low-mass stars	θ_1
0.01–0.08	brown dwarfs	θ_2
0.001–0.01	planetary-mass objects	$\alpha_4 = -0.18$ (fixed)

The parameter θ is restricted to the box $[-2, 0] \times [-2, 10]$. The break masses and the fixed slopes are the defaults of `genulens`, from the fit of Koshimoto et al. [26]. As θ_2 grows, brown dwarfs disappear and the model converges to a limit with no lenses below $0.08\,M_\odot$. Before running the confidence sequences we fixed the upper end of θ_2 as the value at which the distribution of detected timescales is within about 1% total variation of this limit at every θ_1 in the box, which gives 10; the box therefore contains every model that the data could distinguish from the limit.

Generative model. For a field f and parameter θ, `genulens` (version 2.0.0a4) generates a detected event in five steps:

1. a source: a Galactic component c_S (an age bin of the thin disk, the thick disk, the bar, or another component of the model) and a distance D_S, with probability proportional to $s_{c_S}(\theta)\,\sigma_{c_S}(D_S)$;
2. a lens: a component c_L and a distance $D_L < D_S$, with probability proportional to $n_{c_L}(\theta)\,\rho_{c_L}(D_L)$;
3. the lens's initial mass $M \sim \xi_\theta$;

4. given these, with no further dependence on θ: the lens's evolution over its component's age (into a white dwarf, neutron star, or black hole, or not at all), velocities, and the timescale t;

5. an event-rate weight $\Gamma(z)$ and detection with probability $\varepsilon_f(t)$.

Here σ_c and ρ_c are fixed spatial profiles of the fitted Galactic model. The factors $s_c(\theta)$ and $n_c(\theta)$ turn each component's fitted *mass* density into the *number* of source stars and of lenses; they depend on θ because the mean mass of a component's objects, and its split into stars and remnants, are integrals over ξ_θ at the component's age. The parameter therefore enters in three places, not only through the lens mass. Writing z for all latent variables, a detected event has density

$$p_\theta(z \mid f) = \frac{1}{Z_\theta}\, s_{c_S}(\theta)\, \sigma_{c_S}(D_S)\, n_{c_L}(\theta)\, \rho_{c_L}(D_L)\, \xi_\theta(M)\, k(z_{\text{rest}} \mid c_S, c_L, D_S, D_L, M)\, \Gamma(z)\, \varepsilon_f(t), \tag{56}$$

with an unknown normalizing constant Z_θ and a θ-free kernel k for step 4. We write $q_\theta(\cdot \mid f)$ for the induced distribution of t [26]; the measured timescale adds the error of Section H.1. The density of q_θ integrates over distances, masses, and velocities and has no closed form, so the model can be simulated but its likelihood is intractable.

H.3 Exact simulation by reweighting

Why reweight. `genulens` costs a few microseconds per event but about 0.2 s per call, because each call rebuilds the Galaxy for the given θ. The confidence sequence needs fresh references at every candidate and every step, far too many calls. We therefore simulate, once per field, a *bank* of detected events at a proposal $\theta_{\text{prop}} = (-1.2, -1.6)$, and turn bank entries into exact draws at any θ by rejection sampling.

Banks. `genulens` does not draw events in proportion to how often they occur: it returns raw events z with importance weights $\Gamma(z)$. We keep each raw event with probability $\min\{\Gamma(z)/c, 1\}$, $c = 50$, and then with probability $\varepsilon_f(t)$. If $\Gamma \le c$, the kept events are i.i.d. draws from $p_{\theta_{\text{prop}}}(\cdot \mid f)$; otherwise their total variation distance from it is at most $\mathbb{E}[(\Gamma - c)_+]/\mathbb{E}[\Gamma]$. The weights have mean 0.71, and none of the 1.4×10^{10} raw events behind our banks exceeds c, so the kept events are exact draws from the model with weights truncated at c, and the estimate of this bound, computed for every bank, is zero. An entry stores t, the lens's current mass and remnant type, and the components c_L and c_S.

Density ratio. Every factor of p_θ that does not involve θ cancels:

$$\frac{p_\theta(z \mid f)}{p_{\theta_{\text{prop}}}(z \mid f)} = \frac{Z_{\theta_{\text{prop}}}}{Z_\theta}\, w_\theta(z), \qquad w_\theta(z) = \frac{\xi_\theta(M)}{\xi_{\theta_{\text{prop}}}(M)} \cdot \frac{n_{c_L}(\theta)}{n_{c_L}(\theta_{\text{prop}})} \cdot \frac{s_{c_S}(\theta)}{s_{c_S}(\theta_{\text{prop}})}. \tag{57}$$

The weight depends on an entry only through (M, c_L, c_S), and each factor is computed exactly. The mass term uses the gridded density from which `genulens` samples (1,000 log-mass bins, linear within a bin). M is the lens's mass for stars and brown dwarfs and follows from `genulens`'s deterministic initial–final mass relation for white dwarfs; neutron stars and black holes descend from $M > 9\,M_\odot$, where the mass term is constant. The component terms come from `genulens` itself. Because the profiles do not depend on θ, each ratio is the same at every distance and on every sightline (to within 2×10^{-6}), so one call per candidate tabulates them. A component that is switched off in the model has zero density at every θ, is never drawn, and is excluded.

Rejection. Let $B_\theta \ge \sup_z w_\theta(z)$ be the product of the three factors' maxima (the mass term peaks at a grid point, the component terms over the components present). Accepting a bank entry with probability w_θ / B_θ yields an entry with density proportional to $p_{\theta_{\text{prop}}} w_\theta \propto p_\theta$, that is, an exact draw from the model at θ; the constant Z_θ is never needed.

References. For each field, the bank is put in one random order, shared by all candidates together with the uniform draws of the rejection step. Each candidate takes the entries it accepts in that order, K for each event of the field, and never uses an entry twice. Its references are therefore i.i.d. draws from its own model, independent of the observations. Because of this construction, however, a candidate with a lower acceptance probability may reach a given entry earlier in the shared order, so the same entry can be a reference for one

candidate at an early step and for another at a later one. The minimum-evidence alternative is chosen from every candidate's accumulated evidence, so the score at a given step can depend on entries that the true candidate later uses as references, whereas our construction requires the references to be independent of the past information from which the score is chosen. The dependence introduced by simulation-bank reuse is not covered by our conditional-exchangeability argument. We therefore assess coverage empirically for this implementation and do not establish an exact time-uniform guarantee.

Bank sets. We use eleven pairs of independently simulated bank sets. In each pair, one set supplies references (1.2×10^7 raw events per field) and the other supplies simulated observations (6×10^6), so observations never share entries with references. One pair serves the real data; its simulation set also supplies the NRE training data and the KDE score (Section H.5), and is independent of the other ten pairs, which are rotated over the semi-synthetic runs. The rotation matters because a bank is itself a random sample: references drawn from it are exact draws from the model only on average over the bank, and the coverage guarantee includes this randomness.

Validation. At nine values of θ, including all six corners of $\{-2, 0\} \times \{-2, 1.2, 10\}$ and the published fit, and in four fields spanning the survey, we compared reweighted entries of a production bank set with independent direct `genulens` simulations: Kolmogorov–Smirnov tests on $\log t$ and chi-square tests on the lens component, the source component, and the lens class. None of the 144 tests has $p < 0.01$ (the smallest is 0.019), and median timescales agree to 0.9%. With the mass term alone, 35 of 80 comparable tests have $p < 10^{-4}$.

H.4 Model checking

To assess whether the model is well specified, we fit θ on the held-out even seasons and test the fitted model on the odd seasons, which are the ones analysed. The test compares the observed distribution of $\log_{10} \hat{t}_E$ with the one the model predicts: we bin the observed values at their quantiles, simulate the expected count in each bin conditionally on every event's covariates, and compute Pearson's χ^2 statistic. On synthetic data from the model, this statistic follows its nominal χ^2 distribution. Fitting (θ_1, θ_2) on the even seasons gives $(-1.1, 1.0)$, and testing it on the odd seasons gives $p = 0.09$ with 20 bins and $p = 0.19$ with 40 bins, so the data show no evidence against the model. The fitted $\theta_1 \approx -1.1$ agrees with the published -1.13; θ_2 is weakly identified, possibly due to low-cadence fields detecting few short events.

H.5 Confidence sequences and baselines

We use $G = 2{,}048$ scrambled Sobol candidates on the box, $K = 15$ references, $\alpha = 0.1$, the minimum-evidence alternative, and an NRE-C score trained as in Section E.2 on (θ, X, Y) with covariates drawn from the empirical pool of all 3,514 events independently of θ, so that the network estimates the conditional ratio (*conformal NRE-C CS*).

Conformal KDE. Because Y is one-dimensional given X, its conditional density $\hat{p}_\theta(Y_i \mid X_i)$ can be estimated accurately by a kernel density estimate (KDE) from the simulation set of the real-data pair, reweighted to θ, with a kernel width that includes the event's measurement error σ_i. Its log-ratio is a second, near-exact conformal score, which gives the *conformal KDE CS* . Computed from a bank independent of the references and of any bank that generated the observations, it is positive and predictable. The NRE and kernel scores are run together, sharing observations, orderings, and references, so the comparison is paired.

Baselines. Recomputed at every n on the same candidates and orderings, the baseline is the Wilks set $\{\theta : 2[\max_\vartheta \hat{\ell}_n(\vartheta) - \hat{\ell}_n(\theta)] \le \chi^2_{2,0.9}\}$ with $\hat{\ell}_n = \sum_{i \le n} \log \hat{p}_\theta(Y_i \mid X_i)$ from the kernel likelihood (*Wilks KDE CI*); we also compare with the published values of Koshimoto et al. [26].

Semi-synthetic runs. These use the even-season fit $\theta_* = (-1.1, 1.0)$ as the truth. It replaces the nearest of the 2,048 Sobol candidates, so that it is itself a candidate and coverage can be easily assessed. Each repetition takes the covariates of the 1,798 analysed events in a fresh random order and draws each response from $q_{\theta_*}(\cdot \mid X_i)$, with measurement error, using distinct entries of a simulation bank, so the observations are exactly i.i.d. given the covariates. The 300 repetitions are split into ten groups of 30, each with its own independent pair of banks. A repetition miscovers at time n if θ_* has been excluded from the confidence set by then.

Real-data runs. These run the confidence sequences on 200 random orderings of the analysed events. With no known truth, we report θ_1 and θ_2 separately, through the projections of the confidence set onto each (Section C.1.2): for each endpoint of each projection we give its mean over orderings with a normal 95% interval for that mean.

H.6 Results

Semi-synthetic runs. Both the conformal NRE-C CS and the conformal KDE CS keep their empirical cumulative miscoverage below $\alpha = 0.1$ at every n; empty final sets, in 0.7% and 0.3% of repetitions, count as

Table 9: OGLE-IV microlensing, semi-synthetic runs (300 repetitions at $\theta_* = (-1.1, 1.0)$). *Miscoverage*: Empirical cumulative miscoverage over $n \leq 1{,}798$, with a Wilson 95% interval. *Retained*: mean fraction of the candidates still in the set at $n = 1{,}798$. W_{1798} *(θ_1)*: mean width of the projection onto θ_1 at $n = 1{,}798$, over repetitions with a nonempty set (the θ_1 range of the box has width 2). Brackets give Wilson 95% intervals for miscoverage and normal 95% intervals for the means. All conformal sequences share observations and references.

Set	Miscoverage	Retained	W_{1798} (θ_1)
Conformal NRE-C CS (minimum evidence)	0.073 [0.049, 0.109]	0.181 [0.174, 0.188]	0.70 [0.68, 0.72]
Conformal KDE CS (minimum evidence)	0.053 [0.033, 0.085]	0.178 [0.172, 0.184]	0.67 [0.65, 0.69]
Conformal NRE-C CS (oracle alternative)	—	0.109 [0.106, 0.112]	0.49 [0.48, 0.50]
Conformal KDE CS (oracle alternative)	—	0.105 [0.101, 0.108]	0.46 [0.45, 0.47]
Wilks KDE CI, monitored	0.53 [0.47, 0.58]	0.103 [0.100, 0.105]	0.43 [0.42, 0.43]

miscoverage). The oracle versions use θ_* as the alternative, so they can never exclude it, and we report no miscoverage for them. The Wilks KDE CI is valid at a fixed n (miscoverage 0.09 at $n = 1{,}798$) but not under continuous monitoring.

For θ_2, whose true value is 1.0, the confidence sequences show that it is weakly identified: after all 1,798 events they exclude only values below about -1 (median lower end -0.95 with the NRE score), that is, mass functions with many brown dwarfs, but still reach near the top of the box, where brown dwarfs disappear: their upper end exceeds 9 in 96% of repetitions with the NRE score and 97% with the KDE score. We therefore compare the methods' efficiency through their width in θ_1.

For θ_1, NRE-C score is almost as efficient as the near-exact KDE score: the two paired sequences end up with nearly the same width, so learning the likelihood ratio with a network has a low efficiency cost in this experiment. Replacing the minimum-evidence alternative by the oracle shrinks the sequence substantially, showing that most of the extra width comes from estimating the alternative; even with the oracle, the sequence remains slightly wider than the fixed-n Wilks KDE CI.

Real-data runs. After all 1,798 events, the conformal NRE-C CS is nonempty in 99% of the 200 orderings and the conformal KDE CS in all of them. For θ_1, both sequences shrink steadily with n and contain the published value $\theta_1 = -1.13$ [26] in 97% of orderings (and $\theta_2 = -0.18$ in 97.5%). As in the semi-synthetic runs, the NRE-C score is almost as efficient as the KDE score, and both sequences are wider than the Wilks KDE CI, as expected. As in the semi-synthetic runs, the confidence sequences learn little about θ_2: they exclude only low values, whereas the Wilks KDE CI also excludes high ones, including the limit with no brown dwarfs.